Answers

Final answers only, for checking your own work.

Chapter 1 · Exponents & Logarithms

YOUR TURN

1. (a) $3c^5$ (b) $81x^8y^{-12}$ (c) $3a^3b^4$ (d) 1

2. (a) $2x^2 - 4x$ (b) $4a^4 - 3b^2$

3. Proof

4. (a) 9 (b) $\frac{1}{27}$ (c) $\frac{9}{4}$ (d) $\frac{1}{4}$

5. (a) $x^{3/4}$ (b) $x^{-2/3}$ (c) $x^{1/2}$ (d) x

6. (a) 16 (b) 27 (c) 16 (d) 4

7. (a) $3\sqrt{2}$ (b) $5\sqrt{3}$ (c) $6\sqrt{2}$ (d) $\sqrt{3}$

8. (a) $3+\sqrt{5}$ (b) $2\sqrt{3}$ (c) $\frac{\sqrt{7}-\sqrt{5}}{2}$ (d) $7+4\sqrt{3}$

9. 4.7×10^{-5}

10. (a) $320\,000$ (b) 0.000806 (c) $15\,000\,000$ (d) 0.9

11. 7.2×10^{-2}

12. 3.0×10^{5}

13. (a) 5.49×10^4 (b) 6.7×10^{-3} (c) 1×10^7 (d) 1.54×10^{-1}

14. 1.83×10^3

15. (a) 5 (b) -2 (c) 9 (d) 5

16. (a) $\log_3 81 = 4$ (b) $5^3 = x$, so $x = 125$

17. (a) 3 (b) 0 (c) not defined (d) not defined (e) not defined (f) -3

18. (a) x, for all real x (b) x, for $x > 0$ (c) x^2+1 (d) $3x$, for $x > 0$

19. (a) True (b) Only for $x > 0$ (c) True (d) False; $\log_a 1 = 0$

20. (a) 15 (b) 41 (c) 7 (d) $x = 4$ or $x = -1$; both are valid

21. (a) $\log_3 35$ (b) $\log_2 8 = 3$ (c) $\log_a(x^3y)$ (d) $\ln\left(\frac{a^2c}{b^3}\right)$

22. (a) $2\log_5 a - \log_5 b$ (b) $2+p$ (c) $3p+2q$ (d) 2.682 (e) $\frac{\ln 9}{\ln 4}$ (f) $\frac{\log_2 x}{3}$

23. (a) Proof (b) Proof

24. 3.56

25. (a) $x = 3$ (b) $x = \frac{7}{3}$ (c) $x = 6$

26. (a) Proof (b) $\frac{p}{3}$ (c) $\frac{3}{2}$

27. (a) $x = \frac{3}{2}$ (b) $x = 3$ (c) $x = 2$ (d) $x = \frac{1}{3}$

28. (a) 3.32 (b) 1.73 (c) 1.15 (d) 4.84

29. (a) $x = \ln 2$ or $\ln 3$ (b) $x = 1$ or 2 (c) $x = 1$

END-OF-CHAPTER PROBLEMS

1. 4

2. Proof; $x = \log_2 k - \log_2 7$

3. (a) $\frac{2y^{10}}{x^{10}}$ (b) $x = 3$ (c) Proof

4. (a) 81.2 (3 s.f.) (b) 6.04×10^{24} kg

5. Proof; $(\log_2 3)(\log_3 8) = 3$

6. (a) 11 (b) 4 (c) -7

7. (a) $p+q$ (b) $p+\frac{q}{2}$ (c) $2p-2q$

8. (a) Proof (b) $x = 64$

9. Proof; $\log_2 3 \cdot \log_3 5 \cdot \log_5 8 = 3$

10. (a) $2a$ (b) $a/2$ (c) a (d) $3/a$

11. $\frac{7}{4}$

12. $x = 4$

13. $x = 4+4\sqrt{3}$, $y = -4+4\sqrt{3}$

14. $x = 2^{\sqrt{6}} \approx 5.46$ or $x = 2^{-\sqrt{6}} \approx 0.183$

15. $x \approx 2.90$ or $x \approx -0.68$; both lie in the domain

16. (a) $x^2 - 81x - 486 = 0$ (b) $x \approx 86.6$

17. Reduces to $0 = 1$; **no solution**

18. (a) Proof (b) $\frac{1}{2}$

19. $x = 4$ or $x = \sqrt{2}$

20. (a) Power law, valid for $x > 0$ (b) $x = 27$ or $x = \frac{1}{3}$ (c) The left-hand side contains $\log_3 x$, so $x > 0$ is forced before x^2 is considered

21. (a) $x = 8$ (b) $x > 0$, then $x > 1$, then $x > 2$; defined exactly for $x > 2$

22. 5

23. (a) Proof (b) Both sides 0.8451 (4 d.p.)

24. $(x, y) = (3, 9)$ or $(9, 3)$

25. $\frac{\ln 245}{\ln(7/5)}$

26. $x = \ln 5$

27. $x = 1$ or $x = 2$

28. (a) $N(t) = 500 \times 2^{t/3}$ (b) ≈ 19.93 h, that is **19 h 56 min**

29. (a) 250 000 (b) $\approx$ 344 000 (c) $t \approx 14.7$, so **2035**

30. (a) \$32 000 (b) $\approx$ \$14 199 (c) $t \approx 7.16$, so **after 8 years**

31. (a) ≈ 3.46 (b) $\approx 5.01 \times 10^{-9}$ mol/L (c) 10^4; A is **10 000** times more acidic

32. (a) $x > \frac{1}{2}$ (b) $f^{-1}(x) = \dfrac{3^x + 1}{2}$ (c) Proof; a vertical compression by factor $\frac{1}{2}$ and a translation $\frac{1}{2}$ down

33. (a) Proof (b) $\approx 17.7\%$ (c) ≈ 33.2 years

34. (a) 7.14×10^4 times wider (b) 5.5×10^{61} (c) 1.05×10^2 m (d) 42 folds (e) Doubling each fold gives exponential, not linear, growth

35. (a) Proof (b) Proof (c) $x = 16$ (d) $x = \sqrt{2}$

CHALLENGE QUESTIONS

36. (a) 10^{-1} W/m^2 (b) Proof (c) Proof; 3.01 dB (d) 10 sources (e) Proof; a drop of 6.02 dB (f) 15 minutes; 3 dB is a doubling of intensity, so the rule holds intensity $\times$ time constant

37. (a) Proof (b) 302 (c) $0.1072 \in [0, 0.3010)$, so the first digit is 1 (d) Proof (e) $d = 1$: $\approx 30.1\%$; $d = 9$: $\approx 4.6\%$; counts give 9 times (30.0%) for $d = 1$ and 0 for $d = 9$ (f) Counts support the law; 30 is too small a sample

Chapter 2 · Sequences & Series

YOUR TURN

1. (a) 27 (b) -44 (c) 31 (d) -68

2. (a) 4 (b) 8 (c) 3 (d) 6

3. (a) $n = 33$ (b) $n = 15$ (c) $n = 25$ (d) $n = 18$

4. $u_1 = 2$, $d = 3$

5. 101 is the 17th term

6. (a) 222 (b) 1050 (c) 525 (d) 1200

7. (a) $n = 22$, $S_{22} = 1056$ (b) $n = 20$, $S_{20} = 1010$ (c) $n = 13$, $S_{13} = 338$ (d) $n = 39$, $S_{39} = 409.5$

8. 3366

9. $n = 12$

10. 39

11. $u_1 = 3$, $d = 8$, $u_n = 8n - 5$

12. (a) 1.9, 1.7, 2.1, 1.9; close but not equal, so an arithmetic model is reasonable (b) $d \approx 1.9$ cm per week (c) 19.4 cm; week 9 is outside the measured range, so it is extrapolation

13. (a) 486 (b) $\frac{3}{2}$ (c) 4 (d) -384

14. (a) 2 (b) 3 (c) $\pm\frac{1}{3}$ (both real values are valid) (d) 5

15. (a) $n = 9$ (b) $n = 7$ (c) $n = 8$ (d) $n = 9$

16. $u_1 = 2$, $r = 5$

17. 3

18. (a) $\approx$ 30,968 (b) $\approx$ \$58,399 (c) $\approx$ 124 new cases; with $r = 0.8 < 1$ the outbreak dies out

19. (a) 3280 (b) 8.9998 (c) 252 (d) -21

20. 3

21. 6

22. (a) $n = 10$, $S_{10} = 3069$ (b) $n = 6$, $S_6 = 728$ (c) $n = 9$, $S_9 = 171$ (d) $n = 6$, $S_6 = 189$

23. $u_1 = 1$, $r = 2$

24. (a) $\displaystyle\sum_{k=1}^{5} 5k$ (b) $\displaystyle\sum_{k=1}^{5} 2^{k+1}$ (or $\displaystyle\sum_{k=1}^{5} 4 \cdot 2^{k-1}$) (c) $\displaystyle\sum_{k=1}^{10} 3k$ (d) $\displaystyle\sum_{k=1}^{5} k^2$

25. (a) 51 (b) 242 (c) 65

26. $\sum_{k=1}^{n} 5 = 5n$; $\sum_{k=1}^{n} 5k = \frac{5n(n+1)}{2}$

27. (a) 18 (b) 25 (c) $\frac{27}{2}$ (d) $\frac{32}{5}$

28. (a) $u_1 = 16$ (b) $r = \frac{1}{3}$ (c) $u_1 = 9$ (d) $r = -\frac{1}{4}$

29. $\frac{5}{9}$

30. $|r| = 1.5 \geq 1$, so the series diverges and has no sum to infinity

31. $S_\infty = 32$ (for $r = \frac{1}{2}$) or $S_\infty = \frac{32}{3}$ (for $r = -\frac{1}{2}$)

32. (a) $\$3649.96$ (b) $\$9573.44$ (c) $\$716.28$ (d) $\$2439.78$

33. $r = 5.25\%$

34. 9 complete years

35. $\approx \$10\,080$

36. $\approx \$13\,050$

37. (a) 720 (b) 3024 (c) 220 (d) 315

38. (a) 240 (b) 480

39. $4 \times {}^8P_3 = 1344$

40. (a) 55 (b) 165

41. (a) 155 (b) 374 400

42. $n = 10$

43. (a) $x^4 + 12x^3 + 54x^2 + 108x + 81$ (b) $1 + 14x + 84x^2$ (c) $16x^4 - 32x^3 + 24x^2 - 8x + 1$ (d) $1 - 18x + 135x^2$

44. (a) -720 (b) 160 (c) 560 (d) 1512

45. (a) 84 (b) -160 (c) 84 (d) 135

46. $-160x^3y^3$

47. (a) $1-x+x^2-x^3$, $|x| < 1$ (b) $1+2x+3x^2+4x^3$, $|x| < 1$

48. (a) $1 + 2x - 2x^2$, $|x| < \frac{1}{4}$ (b) $1 + \frac{x}{3} - \frac{x^2}{9}$, $|x| < 1$

49. $\frac{1}{3} + \frac{x}{9} + \frac{x^2}{27}$, valid for $|x| < 3$

50. $2 + \frac{x}{12} - \frac{x^2}{288}$, valid for $|x| < 8$

51. $\sqrt{1.1} \approx 1.0488$ (true value 1.048809)

END-OF-CHAPTER PROBLEMS

1. (a) $a + 4d = 17$, $2a + 9d = 31$ (b) $a = 29$, $d = -3$ (c) $u_{11} = -1$

2. (a) $u_1 = 1$, $d = 3$ (b) 590 (c) $n = 21$ (since $S_{21} = 651 > 600$)

3. (a) 3 (b) 455

4. $u_1 = 34$, $d = -2$

5. (a) $u_1 = 4$, $u_2 = 6$ (b) Proof; $d = 2$ (c) $n = 21$

6. (a) $u_{42} = -3$ (b) $n = 40$ or 41 (since $u_{41} = 0$), $S_{40} = S_{41} = 2460$

7. (a) $3n+1$ and $4n-1$ (b) $\mathbf{19}, \mathbf{31}, \mathbf{43}$ (c) Proof; $d = 12$

8. $a = 32$, $r = \frac{1}{2}$

9. (a) Proof; $k = -9$ or $k = 4$ (b) $-\frac{27}{5}$

10. (a) Proof; $d = 6$, $r = 3$ (b) No sum to infinity: $|r| = 3 > 1$

11. (a) $u_2 = 7$, $u_3 = 15$, $u_4 = 31$ (b) Proof; $r = 2$ (c) $2^{n+1} - 1$

12. (a) Proof (b) $d = 2$; 3, 5, 7, 9 (c) $r = 3$ (d) 440 (e) 9840

13. (a) Proof (b) $\frac{58025}{19683}$

14. (a) $n(2n-1)$ (b) $n = 20$

15. (a) $\frac{3}{4}$ (b) Proof (c) 17

16. (a) $2x$ (b) $|2x| < 1 \Rightarrow |x| < \frac{1}{2}$, i.e. $-\frac{1}{2} < x < \frac{1}{2}$ (c) $\frac{1}{1-2x}$

17. $u_1 = 25 - 5\sqrt{15}$, $r = \frac{\sqrt{15}}{5}$

18. $r = 3^{-1/4} \approx 0.760$ or $r = -3^{-1/4} \approx -0.760$; both satisfy $|r| < 1$

19. (a) $u_1 = \frac{142857}{10^6}$, $r = \frac{1}{10^6}$ (b) $\frac{1}{7}$ (c) $1 \div 7 = 0.\overline{142857}$

20. (a) 3.43 m (b) 56.7 m

21. (a) $\$27\,127.31$ (b) 7 years

22. (a) $\$10\,778.81$ (b) $\$9027.08$ (c) Real growth is far smaller than nominal growth

23. (a) $P(1.05)^n$ (b) $\$12\,278.26$ (c) 22.5. After **23** complete years

24. (a) $\$17\,643.78$ (b) 5.87. First below \$10 000: year **6** (c) $\$20\,136.33$

25. (a) A: $3000 + 150n$, B: $3000(1.04)^n$ (b) A: $\$4500$, B: $\$4440.73$ (c) 12

26. (a) Proof (b) $\$6\,243.18$ (c) 12.2. After **13** complete years

27. (a) $P(1.04)^t$ (b) Proof; $r = (1.00325)^{12} \approx 1.03970$ (5 d.p.) (c) Bank A gives more, by $0.00420P$, about 0.42% of P (d) 17.7 years (e) Proof; Bank C leads by $0.01158P$, about 1.16% of P

28. (a) 1260 (b) 1456

29. (a) $\frac{n(n-1)}{2}$. This equals $1+2+\cdots+(n-1)$, the Gauss pairing sum of §2.2: choosing 2 from n counts the same handshakes that the arithmetic sum counts (b) 20

30. (a) ± 2 (b) 448

31. (a) 7920 (b) 1760

32. (a) ${}^{10}C_k 2^{10-k}(-3)^k x^{10-2k}$ (b) $-1\,959\,552$ (c) $-414\,720$ (d) No such term; every power $10-2k$ is even (e) 1 (f) 11 terms; the even powers from x^{10} down to x^{-10}

33. (a) $1-4x+12x^2-32x^3$, valid for $|x| < \frac{1}{2}$ (b) 0.96117 (5 d.p.), matching $\frac{1}{1.02^2}$

34. (a) $2+\frac{x}{4}-\frac{x^2}{64}+\cdots$, valid for $|x| < 4$ (b) $\sqrt{4.2} \approx 2.04938$ (5 d.p.)

CHALLENGE QUESTIONS

35. (a) $\log_2 3$, a constant. Arithmetic (b) $210\log_2 3 \approx 332.84$ (c) 19 (d) $\log_b a$, constant. Arithmetic (e) Proof; $d = \log_b r$ (f) Equal multiplicative steps become equal distances; e.g. decibels

36. (a) $166\,833$ and $100\,500$ (b) Multiples of both are counted twice; $S_{15} = 33\,165$ (c) $234\,168$ (d) $201\,003$, that is $S_3 + S_5 - 2S_{15}$ (e) Proof (f) $104\,104$. Verify with a GDC list, or by summing the two groups separately

37. (a) $1, 0, 1, 0, 1, 0$; the partial sums never settle, so G diverges (b) The bracketings give 0 and 1, so ordinary algebra does not carry over to a divergent series (c) The formula needs $|r| < 1$; here $|r| = 1$ (d) $T_n = \frac{n(n+1)}{2} \to \infty$ (e) It assumes $G = \frac{1}{2}$, a value G does not have, then does arithmetic on quantities that are not numbers (f) $1, \frac{1}{2}, \frac{2}{3}, \frac{1}{2}, \frac{3}{5}, \frac{1}{2}$, approaching $\frac{1}{2}$ (g) $-\frac{1}{12}$ is a value assigned by a different definition of "sum", not the limit of the partial sums; the series still diverges

Chapter 3 · Proofs

YOUR TURN

1. (a) Equation: true only for $x = 4$ (b) Identity: true for every x

2. (a) Proof (b) Proof (c) Proof (d) Proof (e) Proof

3. (a) Proof (b) Proof (c) Proof

4. (a) Proof (b) Proof (c) Proof (d) Proof

5. (a) -3 is an integer and is not positive (b) 2 is prime and even (c) 4 is divisible by 4 but not by 8 (d) $2+3=5$, which is odd (e) 9 is odd and $9 = 3\times 3$ is not prime (f) $a = 1$, $b = -2$

6. (a) $n = 1$ (b) $x = 1$ (any $0 \le x \le 1$) (c) $n = 2$ (d) $a = b = 1$ (e) $p = 11$, since $2^{11} - 1 = 2047 = 23 \times 89$

7. (a) Proof (b) Proof

8. (a) Proof (b) Proof

9. (a) Proof (b) Proof (c) Proof (d) Proof

10. (a) Proof (b) Proof (c) Proof (d) Proof

11. (a) Proof (b) Proof (c) Proof (d) Proof

12. (a) Proof (b) Proof (c) Proof

13. (a) Proof (b) The claim is made for all $n \ge 0$, so the first case checked must be $n = 0$; starting at $n = 1$ would prove the result only for $n \ge 1$ and leave $n = 0$ unestablished

END-OF-CHAPTER PROBLEMS

1. Proof

2. (a) Proof (b) Proof

3. $n = 2$: $n^2+1 = 5$, divisible by 5; $n = 3$: $n^2+1 = 10$, divisible by 5. The claim is false

4. Proof

5. (a) Proof (b) $\frac{1}{5} \times \sqrt{2}$, a non-zero rational times an irrational, so by (a) it is irrational

6. (a) Proof (b) Proof

7. Proof

8. Proof

9. Proof

10. Proof

11. Proof; the base case is $n = 4$ because the claim fails for $n \le 3$ (e.g. $3! = 6 < 8 = 2^3$), while $4! = 24 > 16$

12. Proof

13. Proof

14. Proof

15. Proof

16. Proof

17. Proof

18. Proof

19. Proof

20. Proof

21. (a) $n = 1$: 11 (prime); $n = 2$: 13 (prime) (b) $n = 11$: $n^2 - n + 11 = 121 = 11^2$, not prime

22. (a) Proof (b) Proof

23. (a) Proof (b) Proof

24. Proof

25. Proof

26. (a) Proof (b) $\frac{n(n+1)(n-1)}{3}$ (c) 2660 (d) Proof; a binomial coefficient counts selections, so it is an integer, and the sum is twice an integer, hence even

27. (a) Proof (b) Proof (c) It would need the step $9 \mid a^2 \Rightarrow 9 \mid a$, which is false (e.g. $a = 3$); so no contradiction follows, as it must not, since $\sqrt{9} = 3$ is rational (d) $\sqrt{3} \times \sqrt{3} = 3$, which is rational; irrationality is not preserved by multiplication, so the irrationals are not closed under multiplication

28. (a) $u_2 = \frac{1}{2}$, $u_3 = \frac{1}{3}$, $u_4 = \frac{1}{4}$ (b) $u_n = \frac{1}{n}$ (c) Proof (d) The base case is missing; nothing starts the chain (e) $P(n)$: $n = n + 1$. If $k = k + 1$ then adding 1 gives $k + 1 = k + 2$, so $P(k) \Rightarrow P(k+1)$ for every k; but $n = n + 1$ gives $0 = 1$, so $P(n)$ is false for every n

29. Proof

30. Proof by induction

31. Proof by induction

32. Proof by induction

33. Proof; the limit is $\cos x$

34. (a) Proof: $I_n = \frac{n-1}{n} I_{n-2}$ (b) $I_0 = \frac{\pi}{2}$, $I_1 = 1$, $I_4 = \frac{3\pi}{16}$, $I_5 = \frac{8}{15}$

CHALLENGE QUESTIONS

35. (a) $1, 2, 4, 8, 16$ (b) Proof (c) Proof (d) Proof (e) 31 and 256 (f) In a regular hexagon the three main diagonals meet at the centre, so three interior points coincide and one region is lost: 30

36. (a) $3, 5, 17, 257, 65\,537$, all prime (b) Proof (c) Proof: by the contrapositive of (b), m has no odd factor greater than 1, so m is a power of 2 (d) $F_5 = 4\,294\,967\,297 = 641 \times 6\,700\,417$, so F_5 is composite and Fermat's conjecture is false

37. (a) Take $a = b = \sqrt{2}$ (b) Take $a = \sqrt{2}^{\sqrt{2}}$ and $b = \sqrt{2}$, giving $a^b = 2$ (c) Exactly one of the two cases holds, and each supplies a pair (d) It is non-constructive: it proves a pair exists without saying which

38. (a) Proof (b) Proof (c) An $F_n \times F_{n+1}$ rectangle: its area $F_n F_{n+1}$ equals the total area $F_1^2 + \cdots + F_n^2$ of the squares, which gives the identity

Chapter 4 · Lines & Quadratics

YOUR TURN

1. (a) $x+3 \geq 0 \Rightarrow x \geq -3$ (b) $7-x \geq 0 \Rightarrow x \leq 7$ (c) $x \neq 2$ (d) $x \neq -5$ (e) $x - 4 > 0 \Rightarrow x > 4$ *(strict: $\ln 0$ is undefined)* (f) $2x+1 > 0 \Rightarrow x > -\frac{1}{2}$ (g) $x > 1$ (h) $x^2 - 9 \neq 0 \Rightarrow x \neq \pm 3$

2. (a) Root: $x \geq -2$; denominator: $x \neq 1$. Domain $x \geq -2$, $x \neq 1$ (b) Root: $x \geq 0$; denominator: $x \neq 9$. Domain $x \geq 0$, $x \neq 9$

3. (a) $x^2 \geq 0 \Rightarrow y \geq 1$ (b) $|x| \geq 0 \Rightarrow y \geq -3$ (c) $\sqrt{x} \geq 0$ on $x \geq 0$, so $y \geq 2$ (d) $|x+1| \geq 0$, so $3 - |x+1| \leq 3$; range $y \leq 3$

4. (a) $(x-3)^2 + 2$, so $y \geq 2$ (b) $(x-2)^2 - 4$, so $y \geq -4$ (c) Already in vertex form; $-(x-2)^2 \leq 0$, so $y \leq 5$ (d) $-(x-3)^2 + 5$, so $y \leq 5$

5. (a) -2 (b) 4 (c) 0 (d) $x(x-3) = 0 \Rightarrow x = 0$ or $x = 3$

6. (a) $\frac{x+6}{3}$ (b) $2x - 1$ (c) $\frac{x-7}{2}$ (d) $\frac{1}{x} - 2$, $x \neq 0$ (e) $\frac{3x+1}{x-2}$, $x \neq 2$ (f) $\frac{2x+4}{1-x}$, $x \neq 1$

7. (a) $x \neq 1$ (b) $\frac{x+2}{x-3}$ (c) $x \neq 3$ (d) $f(2) = 8$ and $f^{-1}(8) = 2$

8. (a) No. $x = 4$ gives $y = \pm 2$, so a vertical line meets the graph twice (b) Yes. Each x gives exactly one y

9. Restrict to $x \geq 0$ (or $x \leq 0$); then $f^{-1}(x) = \sqrt{x}$ (or $-\sqrt{x}$), with domain $x \geq 0$. Without the restriction the graph fails the horizontal line test

10. (a) 2 (b) 3 (c) -2 (d) 0; the line is horizontal

11. (a) $2x - 3$ (b) $3x - 5$ (c) $-\frac{1}{2}x + 2$ (d) $\frac{1}{2}x + 5$

12. (a) $-\frac{2}{3}x + 2$; gradient $-\frac{2}{3}$ (b) $2x - 4$; gradient 2

13. (a) $-\frac{1}{4}$ (b) $-3x + 7$, so the gradient is -3 (c) Yes. Both have gradient 2, and the intercepts differ, so the lines are parallel and distinct (d) Yes. $3 \times \left(-\frac{1}{3}\right) = -1$

14. (a) $(3, 2)$ (b) $(2, 1)$ (c) $(3, 1)$ (d) $(3, 2)$

15. (a) $(2, 4)$ (b) $(3, 2)$

16. (a) None. Doubling the first gives $4x + 2y = 8 \neq 9$, so the lines are parallel and distinct (b) None. Equal gradients, different intercepts (c) Infinitely many. The second equation is twice the first, so they are the same line (d) Exactly one. The gradients differ, so the lines meet once, at $(2, 1)$

17. (a) $(1, 2, 3)$ (b) $(2, -1, 3)$ (c) $(3, 1, -2)$ (d) $(1, -2, 4)$

18. (a) Last row is $(0\ 0\ a \mid b)$ with $a \neq 0$: exactly one solution, $(1, 2, 3)$ (b) No solution: the first two equations describe parallel planes (c) Infinitely many solutions: $(4 - t,\ 2,\ t)$, $t \in \mathbb{R}$ (d) $k = 10$: infinitely many solutions, $(4 - 2t,\ t - 1,\ t)$; for every other k the last row reads $0 = k - 10 \neq 0$, so there is no solution

19. (a) $(x+2)^2 - 3$; vertex $(-2, -3)$ (b) $(x-2)^2 - 4$; vertex $(2, -4)$ (c) $(x - 3)^2 + 2$; vertex $(3, 2)$ (d) $(x + 1)^2 + 4$; vertex $(-1, 4)$

20. (a) Vertex $(2, 5)$; axis $x = 2$ (b) Vertex $(4, -13)$; axis $x = 4$ (c) Vertex $(-1, 2)$; axis $x = -1$ (d) Vertex $(3, -1)$; axis $x = 3$

21. (a) $x = 1$ and $x = -5$ (b) -5 (c) $x = -2$ (d) $(-2, -9)$

22. (a) Maximum 1, at $x = 1$ (b) Minimum $\frac{11}{12}$, at $x = \frac{1}{6}$ (c) Minimum -4, at $x = 3$ (d) Maximum 9, at $x = 2$

23. (a) 2 or 3 (b) 4 or -3 (c) -5 or 4 (d) $-\frac{1}{2}$ or 3

24. (a) -1 or -7 (b) $1 \pm \sqrt{6}$ (c) $-3 \pm \sqrt{7}$ (d) $2 \pm \frac{\sqrt{6}}{2}$

25. (a) $-1 \pm \sqrt{2}$ (b) $-3 \pm \sqrt{5}$ (c) 2 or $-\frac{1}{2}$ (d) $\frac{1}{3}$ or -1

26. (a) Factorisation (b) Completing the square, or the formula (c) The quadratic formula (d) Factorisation, as a difference of two squares: $(x - 3)(x + 3)$

27. (a) 0; one repeated real root (b) $-31 < 0$; no real roots (c) $16 > 0$; two distinct real roots (d) $-3 < 0$; no real roots

28. (a) 9 (b) ± 8 (c) $9 - 4k < 0 \Rightarrow k > \frac{9}{4}$ (d) $16 - 4k^2 = 0 \Rightarrow k = \pm 2$; exclude $k = 0$, since then the equation is linear, not quadratic

29. $k < -7$ or $k > 5$

30. (a) $-3 < x < 3$ (b) $(x + 5)(x - 3) < 0 \Rightarrow -5 < x < 3$ (c) $(x - 1)(x - 3) \geq 0 \Rightarrow x \leq 1$ or $x \geq 3$ (d) $-x(x - 4) > 0 \Rightarrow x(x - 4) < 0 \Rightarrow 0 < x < 4$

31. (a) $2x^2 - x - 6 \leq 0 \Rightarrow (2x + 3)(x - 2) \leq 0 \Rightarrow -\frac{3}{2} \leq x \leq 2$ (b) $x^2 - 16 \geq 0 \Rightarrow (x - 4)(x + 4) \geq 0 \Rightarrow x \leq -4$ or $x \geq 4$ (c) All real x: $\Delta = -3 < 0$ and $a = 1 > 0$, so the parabola lies entirely above the axis (d) No solutions: $x^2 \geq 0$, so $x^2 + 4 \geq 4 > 0$ for every x

32. (a) Proof (b) $x^2 + 2x - 15 < 0 \Rightarrow (x + 5)(x - 3) < 0 \Rightarrow -5 < x < 3$. A width must be positive, so $0 < x < 3$

END-OF-CHAPTER PROBLEMS

1. (a) $\frac{x+1}{2}$ (b) Proof

2. (a) x-intercept $\frac{3}{2}$; y-intercept 3 (b) $\frac{3-x}{2}$

3. (a) $x \neq 3$ (b) $\frac{3x+1}{x-2}$

4. $2x - 7$

5. (a) $(1, 6)$ (b) 2 (c) $4\sqrt{5}$

6. (a) x-intercept $(6, 0)$; y-intercept $(0, 4)$ (b) $-\frac{2}{3}x + 4$, gradient $-\frac{2}{3}$

7. (a) $\frac{1}{2}$ (b) $\frac{1}{2}x + \frac{3}{2}$ (c) $-2x + 9$

8. $(1, 2)$

9. $(1, 1, 1)$

10. (a) $k = 7$ (b) $(2+t,\ 1-2t,\ t)$, $t \in \mathbb{R}$ (c) Inconsistent: the three planes have no common point (they form a triangular prism, meeting pairwise in three parallel lines)

11. (a) $(x - 2)(x + 5)$ (b) $x^2 + 3x - 10$

12. $x^2 - 4x + 3$

13. (a) $(x + 3)(x - 1)$; roots -3 and 1 (b) $(-1, -4)$ (c) Upward parabola through $(-3, 0), (1, 0), (0, -3)$, minimum $(-1, -4)$

14. (a) $(x - 3)^2 - 4$ (b) Vertex $(3, -4)$; axis $x = 3$ (c) x-intercepts 1 and 5; y-intercept 5

15. $(x - 4)(x + 2) \leq 0 \Rightarrow -2 \leq x \leq 4$

16. No real roots $\Rightarrow \Delta < 0$: $16 - 4c < 0 \Rightarrow c > 4$

17. $(x + 3)(x - 2) > 0 \Rightarrow x < -3$ or $x > 2$

18. $\Delta > 0$: $k^2 - 16 > 0 \Rightarrow k < -4$ or $k > 4$

19. $k = 6$ or $k = -2$

20. $(t - 3)(t - 7)$, so 3 and 7

21. $b = -6$, $c = 5$

22. Proof; the roots are equal when $k = 1$

23. Proof

24. $(3, 9)$ and $(-1, 1)$

25. (a) $-12 < 0$: no real solution (b) The line does not meet the parabola (it passes below it)

26. (a) 4 s (b) 20 m

27. $k = -\frac{1}{4}$

28. (a) $w(w + 3)$ (b) 5 (reject -8)

29. Proof; maximum area 25

30. Proof

31. (a) Cubic with a local maximum then a local minimum (b) $-2.11,\ 0.254,\ 1.86$ (c) Local maximum $(-1.15,\ 4.08)$; local minimum $(1.15,\ -2.08)$ (d) $-2.60,\ 0.340,\ 2.26$

32. (a) Proof (b) 6.4 m (c) The lorry passes (d) 12.6 m (e) The model is valid only for $|x| \leq 20$

33. (a) Proof (b) $m = 1$ or $m = 9$ (c) $1 < m < 9$ (d) The point of tangency is $(-2, 1)$ (e) $m < 1$ or $m > 9$

34. (a) Last row reduces to $(0\ 0\ a-4 \mid b-7)$, i.e. $(a-4)z = b-7$ (b) $a = 4$ (c) $b = 7$ (d) $(-1, 3, 0) + t(1, -2, 1)$, $t \in \mathbb{R}$ (e) $a \neq 4$: the planes meet at a single point; $a = 4$, $b = 7$: they share a common line; $a = 4$, $b \neq 7$: the last row reads $0 =$ a nonzero number, so there is no solution and the planes have no point in common

CHALLENGE QUESTIONS

35. (a) Proof (b) $y = 4x - 4$ and $y = -4x - 4$, touching at $(2, 4)$ and $(-2, 4)$ (c) Proof (d) $k < 0$ (point below the parabola): two tangents; $k = 0$: one, the x-axis at the vertex; $k > 0$ (point inside): none (e) False: a point on the parabola has exactly one tangent

36. (a) Proof: $\alpha + \beta = S$, $\alpha\beta = P$ (b) Proof; the roots are real exactly when $S^2 - 4P \geq 0$ (c) $x^2 - 6x + 5 = 0$ (d) Proof; for $x^2 - 6x + 5$: $\alpha^2 + \beta^2 = S^2 - 2P = 26$ and $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{S}{P} = \frac{6}{5}$ (e) $P = 1$, with S free; e.g. $x^2 - 5x + 1$, where $S^2 > 4$ and $S^2 - 4$ is not a perfect square

Chapter 5 · Functions & Graphs

YOUR TURN

1. (a) $2x + 3$ (b) $2x + 6$ (c) $x + 6$ (d) $4x$

2. (a) $(x-1)^2$ (b) $x^2 - 1$ (c) 4 (d) 8

3. (a) 49 (b) 9 (c) 4 (d) 14

4. (a) $\frac{1}{x+1}$ (b) $x \neq -1$ (c) $\frac{1}{x} + 1$ (d) $x \neq 0$

5. (a) $\frac{x+2}{3}$ (b) $4x - 4$ (c) $2x + 5$ (d) $\frac{1}{x} + 1$, $x \neq 0$

6. (a) $\frac{x-1}{2}$ (b) Proof (c) Proof (d) Each composite is the identity, which confirms that f^{-1} is the inverse of f

7. (a) $(f \circ g)(x) = x^2 + 4x + 4$; $(g \circ f)(x) = x^2 + 2$ (b) $-\frac{1}{2}$

8. (a) Translation 3 units up (b) Translation 4 units left (c) Reflection in the x-axis (d) Translation 1 unit right

9. (a) Reflection in the y-axis (b) Reflection in the x-axis (c) Vertical stretch, factor 3 (d) Horizontal stretch, factor $\frac{1}{2}$

10. (a) $(5, 6)$ (b) $(2, 15)$ (c) $(1, 5)$ (d) $(2, -3)$

11. (a) $(-1, 3)$ (b) $(-2, -1)$ (c) $(2, 3)$ (d) $(-2, \frac{3}{2})$

12. (a) Translation 3 units left (b) Vertical stretch, factor 2 (c) Translation 3 left, vertical stretch factor 2, translation 1 down (d) Horizontal stretch factor $\frac{1}{2}$, then translation 3 right

13. (a) $y = 3x^2 + 2$ (b) $y = 3x^2 + 6$; the two differ, so the order matters

14. (a) $y = f(4(x-2))$ (b) Horizontal stretch factor $\frac{1}{4}$, then translation 2 units right

15. (a) Even (b) Odd (c) Neither (d) Odd (e) Even (f) Odd

16. (a) Proof (b) Proof (c) Proof (d) $f(f(x)) = 4x \neq x$, so applying f twice does not return the input

17. (a) $x \geq 0$; $f^{-1}(x) = \sqrt{x}$ (b) $x \geq 1$; $f^{-1}(x) = 1 + \sqrt{x}$ (c) $x \geq 0$ (d) x^3 is always increasing, so it is one-to-one on all of $\mathbb{R}$

18. (a) 5, 1 (b) 3, -4 (c) 2, 4 (d) ± 5

19. (a) $x = 2$ (b) $x = 2$ (c) $(0, 4)$ (d) $|-x| = |x|$, so $f(|x|)$ returns the same output for x and $-x$: even

20. (a) -3 or $\frac{1}{3}$ (b) 2 or -1 (c) 2 (the value $\frac{3}{2}$ makes the right side negative and is rejected) (d) No solution: both candidates make the right side negative

21. (a) $1 < x < 7$ (b) $x \leq -7$ or $x \geq 2$ (c) $2 \leq x \leq 4$ (d) $-4 < x < -\sqrt{2}$ or $\sqrt{2} < x < 4$

22. (a) The V with corner $(2, 0)$ meets the line $y = x$ at $(1, 1)$, and lies below it to the right: $x > 1$ (b) $x < -1$ or $x > 3$ (c) $0.347 < x < 1.53$ or $1.88 < x \leq 3$

23. (a) $x = 3$; $y = 0$ (b) $x = -2$; $y = 0$ (c) $x = -1$; $y = 2$ (d) $x = 2$; $y = 1$ (e) $x = 5$; $y = 0$ (f) $x = -1$; $y = 3$

24. (a) Domain $x \neq 5$, range $y \neq 0$; asymptotes $x = 5$ and $y = 0$; y-intercept $(0, -\frac{1}{5})$; no x-intercept, since $\frac{1}{x-5}$ is never 0 (b) Domain $x \neq 0$, range $y \neq 4$; asymptotes $x = 0$ and $y = 4$; x-intercept $(-\frac{1}{4}, 0)$; no y-intercept, since f is undefined at $x = 0$ (c) Domain $x \neq -2$, range $y \neq 1$; asymptotes $x = -2$ and $y = 1$; x-intercept $(1, 0)$; y-intercept $(0, -\frac{1}{2})$ (d) Domain $x \neq 1$, range $y \neq 2$; asymptotes $x = 1$ and $y = 2$; x-intercept $(-\frac{3}{2}, 0)$; y-intercept $(0, -3)$

25. (a) $3 - \frac{9}{x+1}$ (b) $(-1, 3)$ (c) Reciprocal curve centred on $(-1, 3)$, two branches; x-intercept $(2, 0)$, y-intercept $(0, -6)$

26. (a) $x = 1$ and $x = -3$; horizontal $y = 0$ (b) $x = 3$ and $x = -3$; horizontal $y = 0$ (c) $x = 2$ and $x = -2$; horizontal $y = 1$

(d) $x = 2$; oblique $y = x + 2$ (e) $x = -2$; oblique $y = x - 2$

27. (a) $x + \frac{3}{x}$; oblique asymptote $y = x$ (b) $x + 1 + \frac{1}{x-1}$; oblique asymptote $y = x + 1$ (c) $x + 2 + \frac{5}{x-2}$; oblique asymptote $y = x + 2$ (d) $x + 3 - \frac{2}{x+1}$; oblique asymptote $y = x + 3$

28. (a) $x = 0$ (b) $y = x$ (c) Crossing would require $\frac{2}{x} = 0$, which has no solution (d) Two branches: one in the first quadrant above $y = x$, one in the third below it

29. (a) Vertical $x = 4$, horizontal $y = 0$; two branches, no turning point (b) Vertical $x = 1$ and $x = -1$, horizontal $y = 0$; maximum $(0, -1)$; three branches (c) No vertical asymptote, horizontal $y = 0$; maximum $(0, 1)$; one continuous bell-shaped curve above the axis (d) Vertical $x = 1$ and $x = 3$, horizontal $y = 0$; maximum $(2, -1)$; three branches

30. (a) $x = 3$ (b) $(1, \frac{1}{2})$, a maximum (c) $\frac{1}{f(x)} \to 0$: the curve approaches the horizontal asymptote $y = 0$ (d) $f(x) = 1$ or $f(x) = -1$

31. (a) A straight line through $(2, 0)$ of gradient 1, and a reciprocal curve with asymptotes $x = 2$ and $y = 0$ (b) $(3, 1)$ and $(1, -1)$ (c) $2 < x < 3$

32. (a) $-2 < x < 2$ (b) $x \le -3$ or $x \ge 3$ (c) $-2 \le x \le 3$ (d) $x < -1$ or $x > 3$ (e) $1 < x \le \frac{5}{2}$ (f) $-2 < x \le 3$

33. (a) $\frac{1}{4}$ (b) 1 (c) 3 (d) 3 (e) 0 (f) 1

34. (a) $y = 0$; $(0, 5)$ (b) $y = 5$; $(0, 6)$ (c) $y = -1$; $(0, 3)$ (d) $x = 0$; $(1, 0)$ (e) $x = 2$; $(3, 0)$ (f) $x = -5$; $(-4, 0)$

35. (a) Decay; the base 0.5 lies between 0 and 1 (b) Growth; the base $3 > 1$ (c) Decay; the exponent has a negative coefficient, $-0.2 < 0$ (d) Decay; $2^{-x} = \left(\frac{1}{2}\right)^x$ and $\frac{1}{2} < 1$

36. (a) 3 (b) 9 (c) 2 (d) $e^3 \approx 20.1$ (e) $\log_2 \frac{4}{3} \approx 0.415$ (f) 17

37. (a) $e^{x \ln 2}$ (b) $e^{x \ln 5}$

38. (a) $P_0 = 800$; it grows, since $k = 0.03 > 0$ (b) ≈ 1080 (c) ≈ 13.5 years (d) ≈ 15.1 (e) ≈ 3.47 days

39. (a) $x > 2$ (b) $x > -5$ (c) $x > 3$ (d) $x < 4$ (e) $-3 < x < 3$ (f) $x < -1$ or $x > 1$

40. (a) $\ln(x - 2)$, domain $x > 2$ (b) $e^x + 1$, domain $x \in \mathbb{R}$ (c) $\frac{1}{2} \ln(\frac{x-1}{3})$, domain $x > 1$

41. (a) $y = e^x$ is increasing and $y = 3 - x$ is decreasing, so they meet exactly once (b) $x \approx 0.792$ (c) The equation mixes an exponential and a linear term, so it has no closed-form solution; a graph or a numerical method is the only route

42. (a) $90\,^\circ$C (b) $\approx 58.4\,^\circ$C (c) ≈ 10.4 minutes (d) $T \to 20\,^\circ$C, the room temperature; horizontal asymptote $y = 20$

END-OF-CHAPTER PROBLEMS

1. (a) $2x^2 - 3$ (b) $4x^2 - 12x + 9$ (c) ± 2

2. (a) $(1, 6)$ (b) $(4, 4)$ (c) $(1, 8)$ (d) Minimum at $(1, -4)$

3. (a) $(x - 2)^2 + 1$; translation 2 right and 1 up; range $y \ge 1$ (b) $(x - 3)^2 - 9$; translation 3 right and 9 down; vertex $(3, -9)$

4. (a) $\frac{2x+1}{x-1}$, domain $x \ne 1$ (b) $\sqrt{x - 2}$, domain $x \ge 2$ (c) $\ln x$, domain $x > 0$

5. (a) $1 \le x \le 2$ (b) $-2 < x < 3$ (c) $-2 < x \le 3$

6. (a) Odd (b) Even (c) Neither (d) Odd (e) Even (f) Neither

7. (a) Roots $0,\ 2$; vertex $(1, -1)$ (b) Roots 0 and 2, maximum $(1, 1)$, y-intercept $(0, 0)$ (c) x-intercepts ± 1, peak $(0, 1)$, y-intercept $(0, 1)$, arms rising outside $[-1, 1]$

8. (a) $-\frac{1}{2}$ (b) $\frac{x}{1+2x}$

9. (a) $(-1, -1)$, $(1, -2)$, $(2, 2)$ (b) Reflection in the x-axis, then translation 1 right and 2 up; vertex $(1, 2)$

10. (a) $f^{-1}(x) = \frac{x-2}{3}$; $x = -1$ (b) Domain $x \ge 1$, range $y \ge 0$; $g^{-1}(x) = x^2 + 1$, domain $x \ge 0$ (c) $x = 3$ and $y = 0$; $h^{-1}(x) = 3 + \frac{2}{x}$

11. (a) -1 (b) $x = 1$; the point is $(1, 1)$

12. (a) $f(-3) = 7$; $g(-2) = 5$ (b) Proof

13. (a) $a = 3$, $b = 2$; $f^{-1}(x) = \frac{x-2}{3}$ (b) $b = -4$, $c = 3$; translation 2 right and 1 down

14. (a) On $x \ge -1$ the parabola is increasing, so f is one-to-one (b) $-1 + \sqrt{x + 4}$, domain $x \ge -4$

15. (a) $\frac{1}{4}$, $-\frac{3}{2}$ (b) $-\frac{3}{2} \le x \le \frac{1}{4}$

16. $(a, b) = (1, 0)$, or $a = -1$ with any b

17. Proof

18. Proof

19. (a) $(x - 2)^2 - 1$; vertex $(2, -1)$ (b) Roots $1,\ 3$; y-intercept $(0, 3)$ (c) Upward parabola through $(1, 0)$ and $(3, 0)$, minimum $(2, -1)$, y-intercept $(0, 3)$ (d) $y = |f(x)|$: roots $1,\ 3$, maximum $(2, 1)$, y-intercept $(0, 3)$; $y = f(|x|)$: roots $\pm 1,\ \pm 3$, minima $(\pm 2, -1)$, y-intercept $(0, 3)$ (e) $x^2 - 8x + 18$

20. (a) $f(x) = 3 - \frac{7}{x+2}$ and $\frac{7}{x+2}$ is never 0, so the range is $y \in \mathbb{R}$, $y \ne 3$ (b) $\frac{2x+1}{3-x}$, domain $x \ne 3$ (c) $\frac{3x+5}{x+4}$ (d) $x \in \mathbb{R}$, $x \ne -4$ (e) 3

21. (a) Upward parabola, x-intercepts $(\pm 2, 0)$, vertex $(0, -4)$ (b) $x = -2$ and $x = 2$, found by solving $f(x) = 0$, where $\frac{1}{f(x)}$ is undefined (c) $(0, -\frac{1}{4})$; it is the largest value on the interval, hence a maximum (d) $\frac{1}{f(x)} \to 0^+$; horizontal asymptote $y = 0$ (e) Three branches separated by $x = \pm 2$; outer branches positive approaching $y = 0$, middle branch below the axis with maximum $(0, -\frac{1}{4})$

22. (a) $x = -2$ and $y = 0$; $(0, \frac{1}{2})$ (b) $x = 1$ and $y = 2$; $(0, -3)$ (c) $x = -\frac{1}{2}$ and $y = 2$; $(0, -1)$ (d) Reciprocal curve centred on $(1, 2)$, two branches; x-intercept $(-\frac{3}{2}, 0)$, y-intercept $(0, -3)$

23. (a) Domain $x \neq -3$; asymptotes $x = -3$ and $y = 1$; $f^{-1}(x) = \frac{3x+2}{1-x}$ (b) Asymptotes $x = 2$ and $y = 3$; Proof

24. (a) Vertical $x = 0$, oblique $y = x$; x-intercepts ± 1 (b) No asymptote: the graph is the line $y = x + 2$ with a hole at $(1, 3)$, domain $x \neq 1$

25. (a) 500 (b) 4000 (c) 4 hours

26. (a) 80 g; half-life 5 days (b) 20 g (c) \$20000; $\approx$ \$10440 (d) $V(t) \to 0$; the horizontal asymptote $y = 0$

27. (a) $0,\ \ln 2$ (b) 2

28. Both curves pass through $(0, 1)$ with horizontal asymptote $y = 0$; $y = 2^x$ rises to the right, $y = 2^{-x}$ rises to the left

29. Proof

30. (a) $a = 2$ (b) $b = -3$

31. (a) Proof (b) Vertical $x = 2$; oblique $y = x - 2$ (c) y-intercept $(0, -\frac{5}{2})$; no x-intercept (d) Proof (e) Two branches separated by $x = 2$, each approaching $y = x - 2$; left branch below the asymptote through $(0, -\frac{5}{2})$, right branch above it

32. (a) 100 people (b) ≈ 2635 people (c) $t = \frac{\ln 49}{0.8} \approx 4.86$ days (d) $N \to 5000$: the ceiling of the model, the whole population that could ever hear the rumour (e) Proof; gradient -0.8, vertical intercept $\ln 49 \approx 3.89$

33. (a) $x > 2$ (b) $\frac{5}{2}$ (c) $x = 2$; $f(x) \to -\infty$ as $x \to 2^+$ (d) $\frac{e^x+4}{2}$, domain $x \in \mathbb{R}$, range $y > 2$ (e) Translate $y = \ln x$ by 2 right, then $\ln 2$ up

34. (a) Corners at $(-1, 0)$ and $(1, 0)$; the reflected arch peaks at $(0, 1)$ (b) $x = \pm 2$ (c) $-2 < x < 2$ (d) $x = -1$; $x = -3$ is rejected because it makes $3x + 5$ negative

CHALLENGE QUESTIONS

35. (a) Proof (b) Proof (c) Proof (d) Proof (e) e.g. $\frac{3x+1}{x-3}$ and $\frac{x}{x-1}$

36. (a) $k = 0$ or $a = 1$ (b) $h = 0$ or $b = 1$ (c) Proof (d) Two transformations may be applied in either order exactly when they act on different variables, or when one of them is the identity

37. (a) Parabola $y = x^2 - 4$ with the part between $x = -2$ and $x = 2$ reflected above the axis; x-intercepts $(\pm 2, 0)$, local maximum $(0, 4)$ (b) $k = 0$: 2; $k = 3$: 4; $k = 4$: 3; $k = 6$: 2 (c) At $k = 4$ the line passes through the local maximum $(0, 4)$, where two of the four solutions coincide (d) $k = 0$: 2; $0 < k < 4$: 4; $k = 4$: 3; $k > 4$: 2 (e) $k = 0$: 2; $0 < k < -m$: 4; $k = -m$: 3; $k > -m$: 2

38. (a) Proof (b) Proof (c) $k = \frac{1}{10} \ln \frac{7}{4} \approx 0.0560$; half-life $\approx$ 12.4 minutes (d) 22 minutes (e) $e^{-kt} > 0$ for every t, so $T > 20$ always; the horizontal asymptote $y = 20$ carries that fact

Chapter 6 · Polynomials

YOUR TURN

1. (a) Degree 4; lead coefficient 3 (b) Degree 3; lead coefficient -1 (c) Degree 3; lead coefficient 1 (d) Degree 5; lead coefficient -2

2. (a) $x \to +\infty$: $y \to +\infty$; $x \to -\infty$: $y \to -\infty$ (b) Down at both ends (c) Up on the right, down on the left (d) Down at both ends

3. (a) 5 (b) 3 (c) 0 or 2 (d) The degree is odd, so the two ends point in opposite directions; the curve is continuous, so it must cross the axis at least once

4. (a) $2,\ -3,\ 5$; crosses at each (b) $x = 1$ (touches), $x = -4$ (crosses) (c) $x = 0$ (crosses), $x = 3$ (touches) (d) $x = -1$ and $x = 2$; touches at both

5. (a) $x = -1$ with multiplicity 1; $x = 3$ with multiplicity 2 (b) $(0, 9)$ (c) Down on the left, up on the right (d) Cubic crossing at $x = -1$, touching at $x = 3$, through $(0, 9)$, rising to the right

6. (a) $a(x-2)^2(x+1)$ (b) 1 (c) $(x-2)^2(x+1)$

7. (a) $4x^3 - x^2 + 2x - 4$ (b) $x^4 - 2x^3 + 4x^2 - 3$ (c) $x^4 + x^3 + 7x - 3$ (d) $x^3 - 8$

8. (a) Quotient $x + 3$, remainder 0 (b) Quotient $x^2 - 2x$, remainder 0 (c) Quotient $x^2 + 3x + 2$, remainder 3 (d) Quotient x^2+2x+4, remainder 0

9. (a) Quotient $2x^2-3x+3$, remainder -6 (b) Quotient x^2+x-2, remainder 0 (c) Quotient $2x^3+x^2+2x+5$, remainder 4

10. (a) Quotient x^2+1, remainder 0 (b) Quotient x^2-1, remainder 0

11. (a) 2 (b) 0 (c) 2 (d) -5

12. (a) 1 (b) -2

13. (a) $(x-3)$ is a factor of $f(x)$; equivalently $x=3$ is a root (b) 7 (c) $(x+1)$ (d) 0, so $(x-1)$ is a factor

14. (a) $a+b=1$ and $-2a+b=-5$ (b) $a=2$, $b=-1$

15. (a) Yes (b) Yes (c) No (d) Yes

16. (a) $x(x-1)(x+1)$ (b) $x(x-2)(x+2)$ (c) $(x-1)(x+1)(x+2)$ (d) $(x-1)(2x-1)(x+2)$

17. (a) $(x-3)$ (b) $(x-2)^2(x+1)$. The factor $(x-2)$ appears twice

18. (a) $1,\ -2,\ 4$ (b) 3 (twice) and -1 (c) $0,\ 2,\ -2$ (d) $0,\ -2,\ 1$

19. (a) $1,\ 2,\ 3$ (b) $1,\ 2,\ -1$ (c) $1,\ \frac{1}{2},\ -2$

20. (a) $\pm1,\ \pm2$ (b) $x=1$; the quadratic factor has $\Delta=-3<0$, so it contributes no real root

21. (a) Sum 7; product 10 (b) Sum $-\frac{3}{2}$; product $-\frac{5}{2}$ (c) Sum -4; product 4 (d) Sum $\frac{2}{3}$; product $-\frac{8}{3}$

22. (a) Sum 6; product 6 (b) Sum 2; product $\frac{3}{2}$ (c) Sum 3; product -5

23. (a) $x^2-5x+6=0$ (b) Other root 4; $b=-7$ (c) $k=\pm6$ (d) $x^3-5x^2+2x+8=0$

24. (a) $\alpha+\beta=\frac{7}{2}$; $\alpha\beta=3$ (b) $\frac{7}{6}$ (c) $\frac{25}{4}$ (d) $\frac{1}{4}$

25. (a) $\alpha+\beta=-5$; $\alpha\beta=3$ (b) Sum -1; product -3 (c) $x^2+x-3=0$

26. (a) $\alpha+\beta+\gamma=4$; $\alpha\beta\gamma=-6$ (b) 1 (c) 14

27. (a) $-\frac{1}{x-1}+\frac{1}{x-2}$ (b) $\frac{1}{x-2}-\frac{1}{x+3}$ (c) $\frac{1}{x+1}-\frac{1}{x+4}$ (d) $\frac{1}{2(x-1)}+\frac{1}{2(x+1)}$

28. (a) $\frac{1}{x}+\frac{1}{x+1}$ (b) $-\frac{2}{x}+\frac{3}{x-2}$ (c) $\frac{3}{2(x-3)}+\frac{1}{2(x+1)}$ (d) $\frac{4}{3(x-1)}+\frac{11}{3(x+2)}$

29. (a) $\frac{2}{x-2}+\frac{2}{x+2}$ (b) $\frac{8}{3(x-1)}-\frac{5}{3(x+2)}$

30. (a) Numerator and denominator both have degree 2, so the numerator's degree is not lower (b) $1+\frac{1}{x^2-1}$ (c) $1+\frac{1}{2(x-1)}-\frac{1}{2(x+1)}$

END-OF-CHAPTER PROBLEMS

1. (a) Proof (b) $(x-1)(x-3)(x+2)$ (c) $1,\ 3,\ -2$

2. (a) Proof (b) $(x-2)(2x+1)(x+3)$

3. (a) $a=1$, $b=2$ (b) $a=-4$, $b=5$ (c) $c=-13$, $d=18$

4. $k=6$

5. (a) Cubic; roots $x=-1,\ 1,\ 3$; y-intercept $(0,3)$; crosses at each root, rising to the right (b) Cubic; touches at $x=0$, crosses at $x=3$; rising to the right

6. (a) $p=3$, $q=-10$ (b) $p=-5$, $q=2$, $r=8$ (c) x^2-6x+6

7. (a) $(x-1)(x+1)(x+2)$ (b) $1,\ -1,\ -2$

8. (a) $1,\ 2,\ -3$ (b) $1,\ -\frac{1}{2},\ -2$

9. (a) 19 (b) $-\frac{1}{2}$ (c) $\alpha^2+\beta^2=22$; $(\alpha-\beta)^2=8$

10. (a) $(x-1)(x-2)(x-3)=x^3-6x^2+11x-6$ (b) Sum 6; product 6

11. (a) $\pm1,\ \pm2$ (b) Four

12. Proof; the other roots are $-2,\ -3$

13. $k=\pm6$

14. (a) Proof (b) $(x+2)(2x+1)(x-3)$; roots $-2,\ -\frac{1}{2},\ 3$

15. (a) $k=5$ (b) $-1\pm\sqrt{2}$

16. (a) Proof (b) $x=2$; $x^2+1=0$ has no real solution, so there are no other real roots

17. $2x+1$

18. 14

19. Proof

20. Proof

21. $-2<k<2$

22. (a) Proof (b) $(x-3)(2x-1)(x+2)$ (c) $3,\ \frac{1}{2},\ -2$ (d) $(0,6)$; $x\to+\infty$: $f(x)\to+\infty$; $x\to-\infty$: $f(x)\to-\infty$ (e) Cubic rising to the right, crossing at $x=-2,\ \frac{1}{2},\ 3$; y-intercept $(0,6)$ (f) $x\le-2$ or $\frac{1}{2}\le x\le3$

23. (a) Proof (b) $a=-2$, $b=3$ (c) $(x+1)(x^2-3x+6)$ (d) One real root, $x=-1$; the quadratic factor has negative discriminant (e) $4x+4$

24. (a) $\alpha+\beta+\gamma=-p$; $\alpha\beta+\beta\gamma+\gamma\alpha=q$; $\alpha\beta\gamma=-r$ (b) 30 (c) $-\frac{3}{10}$ (d) 2 and 5 (e) $y^3-12y^2+12y+80=0$

25. (a) $\frac{2}{x-1}+\frac{5}{x+2}$ (b) $\frac{1}{4(x-2)}-\frac{1}{4(x+2)}$ (c) $\frac{5}{x}-\frac{4}{x+1}$

CHALLENGE QUESTIONS

26. (a) $S_0 = 2$; $S_1 = p$ (b) Proof (c) $p(p^2 - 3q)$ (d) Proof (e) $S_2 = 7$; $S_3 = 18$; $S_4 = 47$ (f) S_0 and S_1 are integers, and the recurrence builds every later S_k from them using only the integers p and q

27. (a) $P(0) = a_0 = a_n \neq 0$, so $x = 0$ is never a root (b) Proof (c) Proof (d) $x = 1$ (double), $\frac{-3+\sqrt{5}}{2}$, $\frac{-3-\sqrt{5}}{2}$ (e) Proof

28. (a) Proof (b) Proof (c) 1; the partial sums increase towards 1 without reaching it, so $y = 1$ is a horizontal asymptote (d) $\frac{3}{4}$ (e) $k = 3$: $\frac{11}{18}$; $k = 4$: $\frac{25}{48}$; in general $\frac{H_k}{k}$

Chapter 7 · Shapes & Measures

YOUR TURN

1. (a) 5 (b) 7 (c) 5 (d) 5

2. (a) $(3, 3, 2)$ (b) $(4, 1, 2)$

3. (a) $x^2+y^2+z^2 = 25$ (b) $(x-1)^2+(y-2)^2+(z-3)^2 = 16$ (c) $(x-2)^2+(y+1)^2+(z-3)^2 = 36$ (d) $x^2 + y^2 + (z-2)^2 = 7$

4. (a) 5 (b) 13 (c) 67.4°

5. (a) 36π (b) 12π (c) 60 (d) 160π

6. (a) 100π (b) 65π (c) Curved 32π plus flat base 16π: total 48π (d) 80π

7. (a) 10 (b) 60π

8. (a) The cone's curved surface and the hemisphere's curved surface; the two flat circles where they join are inside the solid (b) Cone curved 15π; hemisphere curved 18π: total 33π (c) Cone 12π; hemisphere 18π: total 30π

9. (a) 9.43 (b) 12.9 (c) $C = 75^\circ$; $b \approx 12.2$ (d) 43.5°

10. (a) 7 (b) 12.4 (c) 46.6° (d) 78.5°

11. (a) 10 (b) 27 (c) 20.2

12. (a) 65.4° or 114.6° (b) Proof (c) With the angle included between the two given sides (SAS) the cosine rule gives the third side directly, so no sine of an unknown angle is taken and no supplement arises; SAS fixes the triangle uniquely

13. (a) 22.6° (b) 26.6°

14. (a) 172 m (b) 35.8 m (c) 165 m

15. (a) 270° (b) 300° (c) 245° (d) 020°; adding would give the invalid bearing 380°

16. (a) 5.64 m (b) 2.05 m

17. (a) 90°, so the ship turns through a right angle at Q (b) 64.0 km (c) 081°

18. (a) $\frac{\pi}{3}$ (b) $\frac{3\pi}{4}$ (c) $\frac{7\pi}{6}$ (d) $\frac{\pi}{2}$

19. (a) 135° (b) 120° (c) 57.3°

20. (a) $s = 12$; $A = 48$ (b) $s = 2\pi \approx 6.28$; $A = 6\pi \approx 18.8$ (c) $s = 21.6$; $A = 97.2$ (d) $s = 2\pi \approx 6.28$; $A = 4\pi \approx 12.6$

21. (a) 2.5 rad (b) $\theta = 3$ rad; $s = 18$

22. (a) 64 (b) 29.1 (c) 34.9 (d) 32

END-OF-CHAPTER PROBLEMS

1. (a) 14.1 (b) 45°

2. (a) $AB = 2\sqrt{5}$; $AC = 2\sqrt{5}$; $BC = 4\sqrt{2}$ (b) 78.5°

3. (a) 13 (b) 100π (c) 90π

4. 87π

5. (a) 103.1° (b) 42.9

6. (a) 12.7 (b) 34.9

7. (a) 7.71 (b) 34.7

8. (a) Proof (b) 64.0 km (c) 081°

9. (a) 14.4 cm (b) 86.4 cm^2 (c) 38.4 cm

10. (a) 1.67 rad (b) 10 cm

11. (a) 75 (b) 49.9 (c) 25.1

12. (a) 63.4° or 116.6° (b) 19.3 or 4.85

13. 166

14. (a) 12 (b) 13 (c) 65π

15. 54.6 m

16. (a) 400 (b) 13 (c) 360

17. 36.9° or 143.1°

18. 28.8 km

19. (a) 9 (b) Midpoint $(2, 2, 3.5)$ (c) Proof

20. 0.4 m

21. (a) $\frac{\pi}{6}$ (b) $\frac{5\pi}{4}$ (c) $\frac{5\pi}{3}$

22. 106 m

23. 49.9° or 130.1°

24. (a) 150° (b) 315° (c) $\frac{15\pi}{2} \approx 23.6$

25. Radius $\frac{9}{\pi} \approx 2.86$; height ≈ 11.7

26. $r = 8$, $\theta = 0.5$

27. 16.3

28. $c = 15$ or $c \approx 22.5$

29. (a) Proof (b) 32.8 km (3 s.f.) (c) 080.9°, i.e. 081° (d) 223 km^2 (3 s.f.) (e) 75.8 km

30. (a) 10 cm (b) $5\sqrt{5}$ cm (≈ 11.2 cm) (c) 26.6° (d) 44.3° (e) 236 cm^2

31. (a) 180π cm^3 (≈ 565 cm^3) (b) 4 cm (c) $\frac{160\pi}{3}$ cm^3 (≈ 168 cm^3) (d) $\frac{8}{27}$; volume scales as the cube of the linear scale factor, and $\left(\frac{2}{3}\right)^3 = \frac{8}{27}$, not $\frac{2}{3}$ (e) 11.9 cm (3 s.f.)

CHALLENGE QUESTIONS

32. (a) Proof (b) $a = 5$: none; $a = 8$: two; $a = 12$: one (c) h is the shortest distance from C to the line AB, so a side $a < h$ cannot reach that line and no triangle exists (d) Exactly one triangle; it is right-angled at B (e) Proof; at $a = b$ the second solution gives $B = 180^\circ - A$, so no third angle is left and that triangle is degenerate, leaving only one (f) None for $a \leq b$; exactly one for $a > b$, since an obtuse A forces a to be the longest side

33. (a) Proof (b) Proof (c) Proof (d) $t \approx 0.8165$; $\theta \approx 294^\circ$ (e) Proof; $\frac{h}{r} = \frac{\sqrt{2}}{2}$

34. (a) Proof (b) 1.333×10^7 cubits (c) $33'$: $R \approx 1.415 \times 10^7$, a change of $+6.15\%$; $35'$: $R \approx 1.258 \times 10^7$, a change of -5.63% — one arcminute moves the answer by about 6% (d) Proof (e) R is proportional to θ^{-2}, so a 3% error in θ gives about a 6% error in R; the method is exact but the measurement is fragile, so the result should be quoted to at most two significant figures

Chapter 8 · Trigonometry

YOUR TURN

1. (a) $\frac{1}{2}$ (b) $\frac{\sqrt{2}}{2}$ (c) $\sqrt{3}$ (d) 1

2. (a) $\frac{1}{2}$ (b) $-\frac{1}{2}$ (c) $-\frac{\sqrt{3}}{2}$ (d) $\frac{\sqrt{2}}{2}$

3. (a) First and second (b) Second and third (c) First and third: $\sin$ and $\cos$ share a sign there (d) Fourth only

4. (a) $P(\cos\theta, \sin\theta)$ (b) P lies on a circle of radius 1 centred at the origin, so by the distance formula $\cos^2\theta + \sin^2\theta = 1^2 = 1$ (c) $\tan\theta = \frac{\sin\theta}{\cos\theta}$ and $\cos\frac{\pi}{2} = 0$, so the quotient divides by zero. Geometrically the line OP is vertical and has no gradient

5. (a) $y = x\sqrt{3}$ (b) $\frac{\pi}{4}$ (c) $\frac{3\pi}{4}$

6. (a) 1 (b) $\sin 2\theta$ (c) $\cos 2\theta$ (d) $\cos 2\theta$

7. (a) $\frac{4}{5}$ (b) $\frac{3}{4}$ (c) $\frac{24}{25}$ (d) $\frac{7}{25}$

8. (a) $\frac{4}{5}$, positive in the second quadrant (b) $-\frac{4}{3}$ (c) $-\frac{24}{25}$ (d) $-\frac{7}{25}$

9. (a) $\frac{\sqrt{6}+\sqrt{2}}{4}$ (b) $\frac{\sqrt{6}+\sqrt{2}}{4}$

10. (a) Proof (b) Proof (c) Proof (d) Proof (e) Proof (f) Proof

11. (a) Amplitude 3; period 2π; midline $y = 0$ (b) Amplitude 1; period π; midline $y = 0$ (c) Amplitude 4; period $\frac{2\pi}{3}$; midline $y = -2$ (d) Amplitude 5; period $\frac{\pi}{2}$; midline $y = 3$

12. (a) $-1 \leq y \leq 3$ (b) $-4 \leq y \leq 2$ (c) Amplitude 2, midline -1: $-3 \leq y \leq 1$

13. (a) π (b) $x = \frac{\pi}{2}$ and $x = \frac{3\pi}{2}$ (any $x = \frac{\pi}{2} + n\pi$) (c) $\tan x = \frac{\sin x}{\cos x}$, so it is undefined where $\cos x = 0$; near those values the denominator approaches zero while the numerator does not, so the graph increases without bound

14. (a) Amplitude 20 m; midline 22 m (b) $\frac{\pi}{10}$ (c) $h(t) = 22 - 20\cos\left(\frac{\pi t}{10}\right)$ (d) 22 m, the height of the axle

15. (a) 2 (b) 1 (c) 1 (d) -2 (e) $-\frac{\sqrt{3}}{3}$ (f) -2

16. (a) 1 (b) 1 (c) $\tan\theta$ (d) 1 (e) $\tan\theta$ (f) 2

17. (a) Proof (b) Proof (c) Proof (d) Proof

18. (a) $\frac{\pi}{6}$ (b) $\frac{\pi}{2}$ (c) $-\frac{\pi}{4}$ (d) $\frac{2\pi}{3}$

19. (a) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, $[0, \pi]$, $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ (b) $\frac{\pi}{4}$; not $\frac{3\pi}{4}$, which lies outside the range of $\arcsin$ (c) $\frac{2\pi}{3}$ (d) $-\frac{\pi}{6}$ (e) $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$

20. (a) $\frac{\pi}{3}$ (b) $\frac{3}{2}$ (c) 12 (d) Undefined (e) $\frac{\pi}{4}$ (f) Undefined (g) $\frac{3}{5}$ (h) $\frac{24}{25}$ (i) Undefined (j) $\frac{\pi}{3}$ (k) $\frac{2\sqrt{5}}{5}$ (l) 0.8 (m) $\frac{63}{65}$ (n) $\frac{4\sqrt{5}-3\sqrt{10}}{30}$

21. (a) $\frac{\pi}{6}$, $\frac{5\pi}{6}$ (b) $\frac{\pi}{2}$, $\frac{3\pi}{2}$ (c) $\frac{\pi}{4}$, $\frac{5\pi}{4}$ (d) 0, 2π

22. (a) $\frac{7\pi}{6}$, $\frac{11\pi}{6}$ (b) $\frac{\pi}{6}$, $\frac{11\pi}{6}$ (c) $\frac{5\pi}{4}$, $\frac{7\pi}{4}$ (d) $\frac{2\pi}{3}$, $\frac{5\pi}{3}$

23. (a) 0, $\frac{\pi}{2}$, π, $\frac{3\pi}{2}$, 2π (b) $\frac{\pi}{6}$, $\frac{5\pi}{6}$, $\frac{7\pi}{6}$, $\frac{11\pi}{6}$ (c) $\frac{\pi}{8}$, $\frac{5\pi}{8}$, $\frac{9\pi}{8}$, $\frac{13\pi}{8}$

24. (a) 0, π, 2π, $\frac{\pi}{6}$, $\frac{5\pi}{6}$ (b) $\frac{\pi}{3}$, $\frac{5\pi}{3}$, π (c) $\frac{7\pi}{6}$, $\frac{11\pi}{6}$, $\frac{\pi}{2}$ (d) 0, $\frac{\pi}{3}$, $\frac{5\pi}{3}$, 2π (e) $\frac{\pi}{6}$, $\frac{\pi}{2}$, $\frac{5\pi}{6}$, $\frac{3\pi}{2}$ (f) 0, $\frac{\pi}{4}$, π, $\frac{5\pi}{4}$, 2π

25. (a) $\pm\frac{\pi}{6}$ (b) $-\frac{\pi}{6}$, $-\frac{5\pi}{6}$ (c) $-\frac{\pi}{4}$, $\frac{3\pi}{4}$ (d) $\frac{\pi}{4}$, $\frac{3\pi}{4}$

END-OF-CHAPTER PROBLEMS

1. (a) $\frac{\sqrt{3}}{2}$ (b) $-\frac{1}{2}$ (c) $-\sqrt{3}$ (d) $-\frac{\sqrt{2}}{2}$ (e) $-\frac{1}{2}$ (f) -1

2. (a) $\left(-\frac{\sqrt{3}}{2},\ \frac{1}{2}\right)$ (b) $\frac{5\pi}{4}$ (c) $\sin\theta < 0$, $\cos\theta < 0$, $\tan\theta > 0$ (d) $\frac{\pi}{3}$ and $\frac{2\pi}{3}$; the line $y = \frac{\sqrt{3}}{2}$ cuts the circle at two points, one in each of the quadrants where $y > 0$

3. (a) Proof (b) Proof (c) Proof (d) Proof (e) Proof (f) Proof (g) Proof (h) Proof

4. (a) $\frac{4}{5}$ (positive in the second quadrant) (b) $-\frac{4}{3}$ (c) $-\frac{24}{25}$ (d) $-\frac{7}{25}$

5. (a) $\sin\theta = \frac{2}{\sqrt{5}}$, $\cos\theta = \frac{1}{\sqrt{5}}$ (b) $\frac{4}{5}$ (c) $-\frac{3}{5}$

6. (a) $2+\sqrt{3}$ (b) $\frac{\sqrt{6}+\sqrt{2}}{4}$ (c) $\frac{\sqrt{6}+\sqrt{2}}{4}$ (d) $-2-\sqrt{3}$

7. (a) Proof (b) Proof (c) Proof

8. (a) $\frac{\pi}{6},\ \frac{5\pi}{6}$ (b) $\frac{\pi}{6},\ \frac{7\pi}{6}$ (c) $\frac{\pi}{4},\ \frac{3\pi}{4},\ \frac{5\pi}{4},\ \frac{7\pi}{4}$ (d) $\frac{\pi}{3},\ \frac{2\pi}{3},\ \frac{4\pi}{3},\ \frac{5\pi}{3}$

9. (a) $\frac{\pi}{6},\ \frac{5\pi}{6},\ \frac{3\pi}{2}$ (b) $\frac{\pi}{3},\ \frac{2\pi}{3},\ \frac{4\pi}{3},\ \frac{5\pi}{3}$ (c) $0,\ \frac{2\pi}{3},\ \frac{4\pi}{3},\ 2\pi$ (d) $0,\ \frac{\pi}{3},\ \pi,\ \frac{5\pi}{3},\ 2\pi$

10. (a) $\frac{\pi}{12},\ \frac{5\pi}{12},\ \frac{13\pi}{12},\ \frac{17\pi}{12}$ (b) $\frac{\pi}{6},\ \frac{\pi}{2},\ \frac{5\pi}{6}$ (c) $0,\ \frac{\pi}{2},\ 2\pi$

11. (a) $-\frac{\pi}{6},\ -\frac{5\pi}{6}$ (b) $\pm\frac{\pi}{6}$ (c) $\frac{3\pi}{4},\ -\frac{\pi}{4}$ (d) $\pm\frac{\pi}{4},\ \pm\frac{3\pi}{4}$ (e) $-\frac{5\pi}{6},\ -\frac{2\pi}{3},\ \frac{\pi}{6},\ \frac{\pi}{3}$ (f) $-\frac{11\pi}{12},\ \frac{\pi}{12}$

12. $\frac{\pi}{3}$

13. (a) $\frac{\pi}{4}$ (b) $-\frac{\pi}{6}$ (c) π (d) $\frac{\pi}{6}$, not $\frac{5\pi}{6}$, which lies outside the range of $\arcsin$

14. (a) Amplitude 3; period π (b) Maximum 4, minimum -2

15. $a = 3,\ b = 2,\ d = 4$

16. Maximum 0 at $x = 0,\ 2\pi$; minimum -2 at $x = \pi$; y-intercept 0

17. (a) Maximum 8 m; minimum 2 m (b) 12 hours, the time from one high tide to the next

18. (a) Minimum 2 m; maximum 18 m (b) 30 seconds

19. (a) $R = 5,\ \alpha \approx 0.927$ (b) Maximum value 5

20. $M = 13$ at $x \approx 0.395$

21. (a) 0.565 (b) $\cos(-1.2) = 0.362$ and $\cos(2\pi - 1.2) = 0.362$ (c) Adding 2π returns P to the same point, so its y-coordinate, which is $\sin\theta$, is unchanged (d) Adding π sends P to $(-x,\ -y)$: the tangent $\frac{-y}{-x} = \frac{y}{x}$ is unchanged, but the sine becomes $-y$, so $\sin(\theta+\pi) = -\sin\theta$

22. (a) $\frac{3\pi}{4},\ \frac{5\pi}{4}$; then $\pm\frac{3\pi}{4}$ (b) $-\frac{\pi}{3},\ -\frac{2\pi}{3}$ (c) $-\frac{5\pi}{6},\ -\frac{\pi}{3},\ \frac{\pi}{6},\ \frac{2\pi}{3}$ (d) Cosine is even and $\arccos k \in (0,\pi)$, so the pair is $\pm\arccos k$

23. (a) Maximum 15.6 h at $t = 172$; minimum 8.8 h at $t = 355$ (b) 365 days, one year (c) 8.86 h (d) $t = 81$ and $t \approx 263.5$, the equinoxes (e) $138 \le t \le 207$, about 70 days

24. (a) Proof (b) $-2u^2 + 2u + 1,\ -1 \le u \le 1$ (c) Proof (d) $\frac{\pi}{6},\ \frac{5\pi}{6}$ (e) $0,\ \frac{\pi}{2},\ \pi,\ 2\pi$ (f) $-3 \le f(x) \le \frac{3}{2}$

25. (a) $-\frac{\pi}{6}$ (b) $\frac{\pi}{4}$ (c) $-\frac{\pi}{3}$ (d) $\frac{12}{13}$ (e) $\frac{8}{15}$ (f) $\frac{\sqrt{5}}{5}$ (g) $\frac{\pi}{2}$ (h) Undefined (i) 0 (j) $\frac{3}{4}$

26. (a) Proof (b) Proof (c) Proof (d) Proof (e) Proof (f) Proof (g) Proof (h) Proof (i) Proof (j) Proof

CHALLENGE QUESTIONS

27. (a) Proof (b) Proof (c) $25\sin(\theta + 1.287)$; maximum 25, minimum -25 (d) Proof (e) $x \approx 0.644$; here $k = R$, so $\sin(x+\alpha) = 1$ happens once in the interval

28. (a) Proof (b) $\cos 2\theta = 2c^2 - 1$, $\cos 3\theta = 4c^3 - 3c$ (c) $\cos 4\theta = 8c^4 - 8c^2 + 1$, $\cos 5\theta = 16c^5 - 20c^3 + 5c$ (d) Degree n, leading coefficient 2^{n-1} (e) $\sin n\theta$ is odd in θ for even n; every polynomial in $\cos\theta$ is even

29. (a) 8 m and 4 m; period 12 h; first high tide at 03:00 (b) $1 \le t \le 5$, that is 01:00 to 05:00 (c) 4 hours; the model repeats every cycle (d) Proof (e) 2 h 46 min; $\arccos c$ falls ever more steeply as $c \to 1$, so the window closes at $k = 8$ m

Chapter 9 · Vectors

YOUR TURN

1. (a) 5 (b) 3 (c) 5 (d) $\sqrt{3}$

2. (a) $\begin{pmatrix}1\\1\\7\end{pmatrix}$ (b) $\begin{pmatrix}1\\3\\-1\end{pmatrix}$ (c) $\begin{pmatrix}1\\0\\11\end{pmatrix}$ (d) $\begin{pmatrix}3\\7\\5\end{pmatrix}$

3. (a) $\begin{pmatrix}0\\3/5\\4/5\end{pmatrix}$ (b) $\begin{pmatrix}2/3\\-1/3\\2/3\end{pmatrix}$ (c) $\begin{pmatrix}2\\4\\4\end{pmatrix}$ (d) Dividing by $|\mathbf{v}|$ scales the length to $\frac{|\mathbf{v}|}{|\mathbf{v}|} = 1$; the scalar is positive, so the direction is unchanged

4. (a) Parallel: $\begin{pmatrix}2\\4\\6\end{pmatrix} = 2\begin{pmatrix}1\\2\\3\end{pmatrix}$ (b) Not parallel: the first two components need $k = 2$, but $2 \times 3 = 6 \neq 5$, so no single scalar works

5. (a) $\begin{pmatrix}3\\-2\\4\end{pmatrix}$ (b) $\sqrt{29}$ (c) $\left(\frac{5}{2},\ 1,\ 1\right)$ (d) $\begin{pmatrix}-3\\2\\-4\end{pmatrix}$, the negative of $\overrightarrow{AB}$

6. (a) $(4,0,3)$ (b) $(6,9,3)$ (c) Proof

7. (a) $\overrightarrow{OC} = \mathbf{a} + \mathbf{b}$ (b) Midpoint of OC: $\frac{1}{2}(\mathbf{a} + \mathbf{b})$; midpoint of AB: $\frac{1}{2}(\mathbf{a} + \mathbf{b})$ (c) Proof (d) Proof; MN is parallel to AB and half its length

8. (a) 8 (b) 0 (c) 0 (d) 0

9. (a) 45° (b) 27.3° (c) 60°

10. (a) 6 (b) $k = 8$ (c) $k = 0$ (d) With $\theta = 0$, $\mathbf{a} \cdot \mathbf{a} = |\mathbf{a}||\mathbf{a}| \cos 0 = |\mathbf{a}|^2$; the component form $a_1^2 + a_2^2 + a_3^2$ agrees (e) Parallel, in opposite directions: $|\mathbf{a} \cdot \mathbf{b}| = |\mathbf{a}||\mathbf{b}| = 15$, and the negative sign gives $\theta = \pi$

11. (a) $\mathbf{a} + \mathbf{b}$ and $\mathbf{a} - \mathbf{b}$ (b) $|\mathbf{a}|^2 - |\mathbf{b}|^2$ (c) Proof (d) $|\mathbf{a}| = |\mathbf{b}| = 3$, and $\begin{pmatrix} 3 \\ 3 \\ 4 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} = 0$, so the diagonals are perpendicular

12. (a) $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}$ (b) $\begin{pmatrix} 3 \\ 3 \\ 3 \end{pmatrix}$, or the simpler $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ (c) $\begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

13. (a) $x = 2 + \lambda$, $y = -1 + 3\lambda$, $z = 2\lambda$ (b) $\frac{x-1}{2} = \frac{y-2}{1} = \frac{z-3}{4}$ (c) Yes: $\lambda = 1$ gives all three coordinates

14. (a) 250 km h^{-1} (b) $\begin{pmatrix} 10 \\ 5 \end{pmatrix} + t \begin{pmatrix} 150 \\ 200 \end{pmatrix}$ (c) $\begin{pmatrix} 310 \\ 405 \end{pmatrix}$

15. (a) 10 km h^{-1} (b) $\mathbf{r} = \begin{pmatrix} 2 \\ -1 \end{pmatrix} + t \begin{pmatrix} 6 \\ 8 \end{pmatrix}$ (c) $(5, 3)$ (d) $t = 3$ from x, but then $y = 23 \neq 15$

16. (a) $\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ (b) $\begin{pmatrix} 0 \\ 0 \\ 6 \end{pmatrix}$ (c) $\begin{pmatrix} 4 \\ -2 \\ 1 \end{pmatrix}$ (d) $\begin{pmatrix} -3 \\ 6 \\ -3 \end{pmatrix}$

17. (a) 7.5 (b) $\begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix}$, so the area is 2 (c) 2.29 (d) The angle between $\mathbf{v}$ and itself is 0, so $|\mathbf{v} \times \mathbf{v}| = |\mathbf{v}|^2 \sin 0 = 0$; a vector of zero magnitude is the zero vector

18. (a) $\begin{pmatrix} 5 \\ -5 \\ 5 \end{pmatrix}$ (b) $\mathbf{a} \cdot \begin{pmatrix} 5 \\ -5 \\ 5 \end{pmatrix} = 0$ and $\mathbf{b} \cdot \begin{pmatrix} 5 \\ -5 \\ 5 \end{pmatrix} = 0$, so it is perpendicular to both (c) $\frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$ (or its negative) (d) $\begin{pmatrix} -5 \\ 5 \\ -5 \end{pmatrix}$

19. (a) $x + 2y + 3z = 0$ (b) $\begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$ (or any scalar multiple) (c) $x + y + z = 3$ (d) Yes: $1 + 1 + 1 = 3$

20. (a) $3x + 2y + z = 6$ (b) 80.4°

21. (a) $\lambda = 2$, giving $(5, 2, 0)$ (b) $\begin{pmatrix} 2 \\ 1 \\ -3 \end{pmatrix}$ (c) 11.4°

22. (a) A single point, $(1, 2, 3)$ (b) A line, $\mathbf{r} = \begin{pmatrix} 0 \\ 2 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$ (c) No solution: two parallel distinct planes

23. (a) Foot $(2, 2, 1)$; distance 3 (b) $\sqrt{5}$ (c) Same direction $\begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$ and $(4, 0, 2)$ is not on l, so m is parallel to l; distance 3 (d) From $(5, 2, 4)$ the foot is $(3, 4, 3)$ and the distance is 3, the same as before

24. (a) Foot $(2, 3, 3)$; distance 3 (b) $\frac{14}{3}$ (c) 3 (d) Proof; distance 3, as in part (a)

25. (a) No solution, so the lines do not meet: they are skew (b) $P_1(1 + \lambda,\ \lambda,\ \lambda)$ and $P_2(\mu,\ 2,\ 1 + \mu)$; $\lambda = \mu = 2$ (c) Feet $(3, 2, 2)$ and $(2, 2, 3)$; shortest distance $\sqrt{2}$ (d) $\lambda = 1$, $\mu = 0$ gives $\overrightarrow{P_1 P_2} = \mathbf{0}$, so the distance is 0: the lines intersect at $(2, 3, 3)$

END-OF-CHAPTER PROBLEMS

1. (a) $\begin{pmatrix} 4 \\ 5 \\ 0 \end{pmatrix}$ (b) 3 (c) $\begin{pmatrix} 2/3 \\ -1/3 \\ 2/3 \end{pmatrix}$

2. (a) $\begin{pmatrix} 2 \\ 3 \\ 2 \end{pmatrix}$ (b) $\sqrt{17}$ (c) $(2, \frac{1}{2}, 3)$

3. $-\frac{1}{2}$

4. 60°

5. (a) $(5, 2, 3)$ (b) Yes: P is the point where $\lambda = 2$, so P lies on l

6. 60.5°

7. $\mathbf{a} \times \mathbf{b} = \begin{pmatrix} -2 \\ 5 \\ -4 \end{pmatrix}$, $|\mathbf{a} \times \mathbf{b}| = 3\sqrt{5}$

8. $\frac{\sqrt{3}}{2}$

9. (a) $2x - y + 3z = -3$ (b) At B: $4 - 1 + 0 = 3 \neq -3$, so B does not lie on Π

10. $x + y + z = 1$

11. (a) 5 (b) $\begin{pmatrix} 4/5 \\ -3/5 \\ 0 \end{pmatrix}$ (c) $\begin{pmatrix} 8 \\ -6 \\ 0 \end{pmatrix}$

12. $(6, 9, 3)$

13. $(4, -1, 4)$

14. 27.3°

15. Proof

16. 53.1°

17. $(3, 2, 1)$

18. (a) $\begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}$ (b) $(\frac{4}{3}, \frac{2}{3}, -\frac{4}{3})$

19. (a) 0 (b) 90°

20. 6

21. $\begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$

22. 70.5°

23. $\mathbf{a} \times \mathbf{b} = \begin{pmatrix} 20 \\ -15 \\ 0 \end{pmatrix}$, $|\mathbf{a} \times \mathbf{b}| = 25$

24. -2

25. (a) $\begin{pmatrix} 2 \\ -1 \\ 10 \end{pmatrix} + t \begin{pmatrix} 3 \\ 4 \\ 0 \end{pmatrix}$ (b) 5 m s^{-1} (c) Yes, at $t = 5$ s

26. $\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$

27. (a) $\begin{pmatrix}2\\1\\0\end{pmatrix} + t\begin{pmatrix}1\\0\\-1\end{pmatrix}$ (b) Proof (c) A triangular prism; the planes have no common point

28. Proof; the lines intersect at $(2, 3, 3)$

29. 2

30. Unit vectors: $\pm\frac{1}{3\sqrt{5}}\begin{pmatrix}-2\\5\\-4\end{pmatrix}$

31. (a) $6x + 3y + 2z = 6$ (b) $\frac{7}{2}$

32. 26.4°

33. (a) $\overrightarrow{AB} = \begin{pmatrix}-1\\3\\1\end{pmatrix}$, $\overrightarrow{AC} = \begin{pmatrix}2\\1\\-1\end{pmatrix}$ (b) $\begin{pmatrix}-4\\1\\-7\end{pmatrix}$ (c) $\frac{\sqrt{66}}{2} \approx 4.06$ (d) $4x - y + 7z = 15$ (e) $\frac{15}{\sqrt{66}} = \frac{5\sqrt{66}}{22} \approx 1.85$ (f) $\frac{5}{2}$

34. (a) Proof (b) They intersect, at $\left(\frac{11}{3}, \frac{2}{3}, \frac{13}{3}\right)$ (c) 90° (d) $y + z = 5$ (e) The lines intersect; skew lines lie in no plane

35. (a) Both 500 km h^{-1} (b) $\begin{pmatrix}600-300t\\100t\\0\end{pmatrix}$ (c) Proof (d) $t = 1.8$ h, least distance $\sqrt{36\,000} \approx 190$ km

36. (a) $\mathbf{r} = \begin{pmatrix}-2\\3\\0\end{pmatrix} + \lambda\begin{pmatrix}1\\-2\\1\end{pmatrix}$ (b) One point, $(m-3,\ 5-2m,\ m-1)$; with $m = 5$ that is $(2, -5, 4)$ (c) $k = 1$ and $m = 1$ (d) $k = 1$, $m \neq 1$: Π_3 parallel to Π_1, so no common point

37. (a) $\begin{pmatrix}2\\-1\\1\end{pmatrix}$, $\begin{pmatrix}1\\2\\2\end{pmatrix}$ (b) $\begin{pmatrix}-4\\-3\\5\end{pmatrix}$ (c) $\frac{5\sqrt{2}}{2} \approx 3.54$ (d) $4x + 3y - 5z = 5$ (e) It is perpendicular to two vectors lying in the plane

38. (a) 13 and 7 m s^{-1} (b) They cross at $(9, -4, 24)$ (c) Reached at $t = 2$ and $t = 3$, so not at the same moment (d) 7 m (e) $\approx 38.7^\circ$

CHALLENGE QUESTIONS

39. (a) The perpendicular is the shortest segment from P to l (b) Proof (c) $\lambda = \frac{5}{3}$, $F\left(\frac{13}{3}, \frac{5}{3}, \frac{1}{3}\right)$, distance $\frac{\sqrt{21}}{3} \approx 1.53$ (d) $\frac{\sqrt{21}}{3} \approx 1.53$, the same answer (e) Proof

40. (a) Proof; the third component would need $4 = 3$, so no point lies on both and the lines are skew (b) $\mathbf{d}_1 \times \mathbf{d}_2$ is the only direction perpendicular to both (c) Proof (d) $\frac{2}{\sqrt{14}} = \frac{\sqrt{14}}{7} \approx 0.535$ (e) It gives zero, which is correct: intersecting lines are zero distance apart

41. (a) $\mathbf{v} \times \mathbf{w} = \begin{pmatrix}1\\4\\-2\end{pmatrix}$, so $\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) = 9$ (b) Proof (c) $\mathbf{v} \cdot (\mathbf{w} \times \mathbf{u}) = 9$, unchanged by a cyclic rotation; $\mathbf{u} \cdot (\mathbf{w} \times \mathbf{v}) = -9$, since swapping two reverses the sign. The magnitude, and so the volume, is the same in every case (d) Coplanar vectors make $\mathbf{v} \times \mathbf{w}$ perpendicular to $\mathbf{u}$, so the triple product is zero; here it is 0, so the three vectors are coplanar (e) $\frac{3}{2}$

Chapter 10 · Statistics

YOUR TURN

1. (a) Discrete: it can only take whole-number values (b) Continuous: it can take any value in a range (c) Discrete (d) Continuous

2. (a) 20 males (b) 30 females (c) Each group appears in proportion to the population

3. (a) 50 (b) Systematic sampling (c) Gym users exercise more than the general population (d) 40 (e) Quota sampling; the choice within each quota is not random, so it can be biased

4. (a) 10 (b) 9 (c) 8 (d) 10

5. (a) 10.5 (b) 7, which occurs three times (c) 9 (d) 16.5

6. (a) 12 (b) 21

7. (a) 22 (b) 63 (c) 2.86 (d) 3, the value with the highest frequency

8. (a) $5,\ 15,\ 25,\ 35$ (b) 19.3 (c) $10 \leq x < 20$ (frequency 10) (d) Each value is assumed to lie at its class midpoint

9. (a) 18 (b) 5 (c) 14 (d) 9

10. (a) $\bar{x} = 10$, $\sigma \approx 3.06$ (b) $\bar{x} \approx 26.7$, $\sigma \approx 8.82$ (c) 3.37

11. (a) Mean 150; standard deviation 18 (b) Mean 40; standard deviation unchanged at 5 (c) Mean 50; standard deviation 6 (d) Mean 75; standard deviation 18. The -5 does not affect the spread

12. (a) All distances between values are unchanged (b) The IQR ignores the extreme quarters; the range does not

13. (a) 10 (b) $Q_1 = 6.5$, $Q_3 = 14$ (c) $4,\ 6.5,\ 10,\ 14,\ 18$ (d) 7.5

14. (a) Upper boundary $= 80$; since $95 > 80$, it **is** an outlier (b) Lower boundary $= 0$; since $2 > 0$, it is **not** an outlier (c) Lower boundary -6; upper boundary 34

15. (a) 12 (b) $20,\ 40,\ 60$ (c) Histograms show continuous data; bar charts show separate categories

16. (a) Strong negative linear correlation (b) Very weak, or effectively no, linear correlation (c) Very strong positive linear correlation (d) Moderate negative linear correlation

17. (a) 0.936 (b) 0.989

18. (a) r measures *linear* association only; on $-3 \leq x \leq 3$ the falling and rising halves cancel, so $r \approx 0$ (b) Age is a third variable raising both; correlation shows association, not cause

19. (a) $1.7x + 0.9$ (b) $1.45x + 2.3$

20. (a) 19 (b) 18 (c) 40 lies far outside the range $2 \le x \le 15$ used to fit the line. There is no evidence the relationship continues that far, so the prediction is extrapolation and unreliable

21. (a) $1.5x + 8$ (b) The x on y line; predicting x from y reverses the roles, so the line minimising horizontal distances is needed

END-OF-CHAPTER PROBLEMS

1. (a) 23.25 min (b) $20 \le t < 30$ (c) $20 \le t < 30$

2. (a) Five-number summary: $12,\ 18.5,\ 23,\ 27.5,\ 46$ (b) Upper boundary $= 41$; since $46 > 41$, it is an outlier (c) Positively skewed: a long right tail, with the maximum far above Q_3

3. (a) 0.991. A very strong positive linear correlation: more revision is associated with higher scores (b) $2.30x + 5.62$ (3 s.f.) (c) Gradient: each extra hour of revision is associated with about 2.3 more marks. Intercept: a student who does no revision is predicted to score about 5.6 marks, though $x = 0$ lies below the data range (d) 21.7, so about 22 (e) 25 is far outside the data range ($2 \le x \le 12$), so this would be unreliable extrapolation

4. (a) Mean 57; standard deviation unchanged at 8 (b) Mean 67.5; standard deviation 12 (c) Adding a constant shifts the centre but not the spread

5. (a) A: 20, B: 50, C: 30 (b) Each town is represented in proportion to its size

6. (a) Plot $(10, 4)$, $(20, 16)$, $(30, 34)$, $(40, 52)$, $(50, 60)$ and join with a smooth increasing curve (b) Median ≈ 27.8; IQR ≈ 16.9. (Accept reasonable readings.) (c) 34.4

7. 67.8

8. 11

9. (a) Same mean volume, but Y's fills vary far more (b) Machine X, with the smaller standard deviation

10. (a) $0.61x+7.7$ (b) \$29.1 (c) $x = 1.63y-12.4$, giving 36.5 m^2

11. (a) Mean ≈ 14.1; median 8 (b) Mean ≈ 7.33; median 7.5 (c) The mean, because it uses every value; the median barely moves, as it depends only on position

12. 79

13. (a) $Q_1 = 55$, $Q_3 = 63$, IQR $= 8$ (b) Proof (c) The median: the outlier 90 inflates the mean

14. (a) $-2x + 31$ (b) 23

15. (a) 17 (b) Proof (c) B has the higher median and the smaller spread (d) Yes, possibly: the summary is symmetric

16. (a) Proof (b) $10 \le t < 20$ (frequency 18) (c) $10 \le t < 20$ (d) 9.41 min

17. (a) $a = 0.8$, $b = 20.4$ (b) 80.4 (c) 69.2

18. (a) 0.998 (3 s.f.). A very strong positive linear correlation: hotter days are associated with more sales (b) $7.06x - 15.7$ (3 s.f.) (c) About 7 more ice creams sold per extra degree (d) 154 ice creams (e) 40 °C lies far outside the data range

19. (a) $\bar{x} = 7$, $\bar{y} = 15$ (b) 9 (c) Predicting x requires the x on y line

20. (a) Systematic (b) Stratified (c) Convenience (d) Monday-morning shoppers are not representative of all customers (e) 20 juniors, 30 seniors

21. 15

22. (a) -0.92: strong negative; -0.35: weak negative; 0.08: no linear correlation; 0.85: strong positive (b) Correlation is not causation; a third variable is physical inactivity (c) No effect: converting cm to m is a linear rescaling, and r is unchanged by a linear change of scale

23. (a) Five-number summary: $5.4,\ 8.55,\ 16.15,\ 22.6,\ 26.9$ (b) Proof (c) $\bar{x} = 15.83$ °C, agreeing with the published value (d) Roughly symmetric; it shows variation between months, not between days

24. (a) Proof (b) Proof; equality only when $\sigma = 0$, that is when all values are already equal

25. Sets: $\{4, 4, 7, 8, 17\}$, $\{4, 4, 7, 9, 16\}$, $\{4, 4, 7, 10, 15\}$, $\{4, 4, 7, 11, 14\}$, $\{4, 4, 7, 12, 13\}$

26. (a) $\bar{x} = 7.1875$, $\bar{y} = 8.75$ (b) 0.917

27. Proof; the constant cancels in every deviation, leaving the spread unchanged

28. (a) $k = 20$; random start in 1–20, then every 20th (b) Quicker to carry out, and spreads the sample evenly through the register (c) 20 is a multiple of the period 4, so every pick has the same family position; with start 2, all 45 are mothers (d) 18 juniors, 27 seniors (e) Non-response bias; follow up the 27 non-responders

29. (a) Convenience (self-selected); shop visitors cycle far more than average (b) Systematic; 60 (c) $75,\ 120,\ 70,\ 35$ (d) Cycling distance varies sharply with age, so each band is guaranteed its true proportion (e) The fault is bias in the method, not sample size

30. (a) Quota (b) Selection within each group is not random (c) Only library users can be sampled, yet non-users are affected by the hours (d) Simple random or stratified sample from the full student register

CHALLENGE QUESTIONS

31. (a) Mean 7; median 7 (b) Mean $= \frac{24+k}{5}$, median $= 7$ (c) 76 (d) The three smallest values are unaffected and $k \geq 9$ stays last, so the third ordered value is always 7 (e) Raising one value by D moves the mean by $\frac{D}{n}$, which is unbounded, while the median can pass at most one neighbour. So the median is resistant; the mean is not (f) $\lceil n/2 \rceil$ values; for $n = 5$ that is three

32. (a) Proof (b) 54 (c) Proof (d) 6.10 (e) Proof; equality exactly when $\bar{x}_1 = \bar{x}_2$. Combining two groups is at least as variable as the groups themselves, and the further apart their means, the greater the extra spread

33. (a) Proof (b) Proof (c) $-1.5,\ -0.5,\ 0,\ 0.5,\ 1.5$; their mean is 0 and their standard deviation is 1, agreeing with (a) and (b) (d) Student P ($z = 1.6$ versus $z = 1.125$) (e) Standardising removes differences in mean and spread between the tests

Chapter 11 · Probability

YOUR TURN

1. (a) $\frac{1}{2}$ (b) $\frac{1}{4}$ (c) $\frac{2}{5}$ (d) $\frac{3}{10}$ (e) $\frac{1}{4}$

2. (a) 0.65 (b) 15 (c) $\frac{3}{4}$

3. (a) $\{HH,\ HT,\ TH,\ TT\}$; exactly one head has probability $\frac{1}{2}$ (b) 0.24 (c) 60 (d) The coins are fair and symmetric, so the probability can be calculated; the spinner is not, so it can only be estimated from data

4. (a) $\frac{7}{12}$ (b) 0.82 (c) $\frac{15}{16}$

5. (a) 84 (order does not matter) (b) 504 (order matters) (c) 120 (order matters) (d) $2\,598\,960$ (order does not matter)

6. (a) 252 (b) $\frac{10}{21}$ (c) $\frac{1}{42}$ (d) $\frac{41}{42}$

7. (a) $\frac{33}{91} \approx 0.363$ (b) $\frac{58}{91} \approx 0.637$ (c) $\frac{66}{455} \approx 0.145$

8. (a) 0.7 (b) 0.9 (c) 0.1 (d) $P(A \cap B) = 0.3 + 0.4 - 0.7 = 0$, so they are mutually exclusive

9. (a) $\frac{11}{26}$ (b) $\frac{7}{8}$ (c) $\frac{3}{5}$ (d) $n(A \cap B') = 15$, $n(A' \cap B') = 17$ (e) Exactly one: $\frac{2}{5}$; none: $\frac{7}{30}$

10. (a) $\frac{1}{6}$ (b) $\frac{5}{18}$ (c) $\frac{1}{6}$

11. (a) $\frac{3}{20}$ (b) $\frac{7}{16}$ (c) $\frac{47}{80}$

12. (a) 0.4 (b) 0.18 (c) 0.28 (d) $\frac{1}{6}$

13. (a) 0.2 (b) $0.15 \neq 0.2$, so **not** independent (c) $\frac{1}{221}$

14. (a) $\frac{2}{5}$ (b) $\frac{3}{5}$ (c) $P(S)P(A) = \frac{6}{25} = P(S \cap A)$, so they are independent (d) $\frac{1}{3}$

15. (a) $\frac{9}{25}$ (b) $\frac{3}{10}$ (c) $\frac{8}{15}$

16. (a) 0.81 (b) 0.25 (c) 0.421 (d) 0.410

17. (a) $\frac{1}{6}$ (b) $\frac{3}{10}$ (c) 0.815 (d) 0.185

18. (a) 0.72 (b) 0.162 (c) 0.625

19. (a) 0.885 (b) $\frac{2}{3}$ (c) 0.862

20. (a) 0.022 (b) 0.409 (c) 0.227 (d) A has the lowest defect rate, 1%: its contribution $0.5 \times 0.01 = 0.005$ is smaller than 0.009 and 0.008, so the large share is more than cancelled by the low rate

END-OF-CHAPTER PROBLEMS

1. (a) 8 (b) $\frac{6}{25}$ (c) $\frac{2}{7}$

2. (a) 0.3 (b) 0.48 (c) $0.30 \neq 0.27$, so they are **not** independent

3. (a) Tree diagram: $\frac{7}{10}, \frac{3}{10}$; then $\frac{6}{9}, \frac{3}{9}$ and $\frac{7}{9}, \frac{2}{9}$ (b) $\frac{7}{15}$ (c) $\frac{8}{15}$ (d) $\frac{8}{15}$

4. (a) 0.1875 (b) 0.24 (c) Most positive results are false alarms

5. (a) 0.65 (b) 0.8 (c) Independence needs $P(A \cap B) = 0.15$, mutual exclusivity needs $P(A \cap B) = 0$. Both have non-zero probability, so the two conditions contradict each other

6. (a) $\frac{1}{6}$ (b) $\frac{11}{36}$ (c) $\frac{2}{11}$

7. (a) 0.034 (b) 0.441

8. 0.5

9. (a) 0.4096 (b) 0.5904

10. 0.417

11. (a) 7 (b) $\frac{7}{40}$ (c) $\frac{4}{5}$

12. (a) 0.38 (b) 0.6 (c) 9.6

13. (a) 0.10 (b) 0.35 (c) 0.25 (d) $P(A)P(B) = 0.18 \neq 0.10 = P(A \cap B)$, so **not** independent

14. (a) A 4×6 grid of the 24 equally likely outcomes (b) $\frac{1}{8}$ (c) $\frac{1}{4}$ (d) $\frac{1}{3}$

15. (a) Tree diagram: $0.85,\ 0.15$; then $0.7,\ 0.3$ and $0.4,\ 0.6$ (b) 0.655 (c) 0.908

16. (a) $\frac{1}{5}$ (b) $\frac{3}{5}$ (c) $\frac{4}{5}$

17. (a) $P(A \mid B) = 0.6 = P(A)$: knowing B leaves $P(A)$ unchanged, so A and B are independent (b) 0.6 (c) 0.6; A' and B are independent too

18. (a) Rows Boy/Girl, columns Left/Right: $10, 30, 40$; $6, 34, 40$; totals $16, 64, 80$ (b) $\frac{17}{40}$ (c) $\frac{5}{8}$ (d) $P(L \mid B) = 0.25 \neq 0.2 = P(L)$, so **not** independent

19. (a) 0.059 (b) 0.508

20. (a) 0.15 (b) 0.75 (c) $\frac{6}{11}$

21. (a) 0.081 (b) 0.410 (c) 29

22. (a) $\{BB,\ BG,\ GB,\ GG\}$, equally likely (elder listed first) (b) $\frac{1}{3}$ (c) $\frac{1}{2}$ (d) The conditioning events differ: three outcomes against two

23. (a) 80 (b) 20 (c) 25

24. (a) 0.167 (b) 0.798 (c) It raises the probability from about 17% to 80%

25. Proof

26. $\frac{5}{14}$

27. (a) Venn diagram: cycle only 90, both 40, train only 50, neither 20 (b) 0.1 (c) $\frac{4}{9}$ (d) $P(C)P(T) = 0.2925 \neq 0.2 = P(C \cap T)$, so **not** independent (e) 0.00955 (f) Larger: with replacement the second probability stays at $\frac{20}{200}$ instead of falling to $\frac{19}{199}$, and $0.01 > 0.00955$

28. (a) Tree diagram: $0.06,\ 0.94$; then $0.95,\ 0.05$ and $0.02,\ 0.98$ (b) 0.0758 (c) 0.248 (d) 0.00325 (e) 15 (f) Well for one customer, but about 15 defectives still pass

29. (a) 0.7 (b) 0.233 (c) 0.992 (d) Only 3 batteries are flat; if the first three tested are all flat, the fourth must work (e) $P(N=3) = \frac{7}{120}$, $P(N=4) = \frac{1}{120}$; sum $= \frac{7}{10} + \frac{7}{30} + \frac{7}{120} + \frac{1}{120} = 1$

30. (a) $658\,008$ (b) $\frac{1}{658\,008} \approx 1.52 \times 10^{-6}$ (c) $\frac{2975}{329\,004} \approx 0.00904$ (d) The sample space nearly doubles to $1\,221\,759$, so the jackpot chance roughly halves; rarer jackpots roll over and grow, which is what attracts players

31. (a) 126 (b) $\frac{10}{21} \approx 0.476$ (c) $\frac{20}{21} \approx 0.952$ (d) Directly it is three cases (1–3, 2–2, 3–1); the complement is two, and each is a single combination

32. (a) Proof; 3360 (b) $\frac{1}{8}$ (c) $\frac{3}{28} \approx 0.107$

CHALLENGE QUESTIONS

33. (a) $\frac{364}{365}$ (b) The second must differ from the first ($\frac{364}{365}$) and the third from both ($\frac{363}{365}$); the stages multiply (c) $\prod_{k=1}^{n-1} \frac{365-k}{365}$ (d) $n = 10$: 0.117; $n = 23$: 0.507; $n = 50$: 0.970 (3 d.p.) (e) 23, where it is 0.5073 (f) The guess fixes one person and asks who matches them; the event is that *some* pair matches, and 23 people give ${}^{23}C_2 = 253$ pairs (g) $253 \times \frac{1}{365} \approx 0.693$, close to 1, which is why a match is near even money at $n = 23$ (h) 119 students, where a clash has probability 0.506; an identifier space must be sized against the number of *pairs* of users, so it has to be far larger than the population

34. (a) $\frac{1}{3}$ (b) $\frac{2}{3}$ (c) $\frac{1}{3}$. Switching is better: it wins twice as often (d) $\frac{1}{2}$, 1, 0; then $P(\text{prize behind } 2 \mid \text{opens } 3) = \frac{2}{3}$ (e) $\frac{1}{2}$, $\frac{1}{2}$, 0; then $P = \frac{1}{2}$, so switching gains nothing (f) The knowing host opens door 3 with certainty when the prize is behind 2 but only half the time when it is behind 1; that asymmetry is the information. The ignorant host's probability is $\frac{1}{2}$ either way, so nothing shifts (g) Proof

Chapter 12 · Probability Distributions

YOUR TURN

1. (a) 0.3 (b) $k = 0.25$, $P(X \leq 1) = 0.40$ (c) 0.1 (d) $\frac{3}{10}$ (e) $P(X > 1) = 0.5$, $P(1 \leq X \leq 2) = 0.7$

2. (a) All three probabilities are positive and sum to 1, so it is a distribution; $P(X \geq 2) = \frac{13}{18}$ (b) $a = 0.3$, $b = 0.3$ (c) 0.6 (d) $\frac{1}{14}$

3. (a) 2.3 (b) 1.6 (c) 0.5, an expected gain of \$0.50 (d) Expected loss of \$0.25; the game is not fair

4. (a) $E(X^2) = 3.4$, variance 0.84 (b) $E(3X - 1) = 11$, variance 18

5. (a) Expected win \$1.40; a fair stake is \$1.40 (b) $-\$0.60$ per spin; an expected loss of \$30 over 50 spins (c) $E(2X+1) = 7$, variance 20, $E(X^2) = 14$

6. (a) 0.273 (b) 0.233 (c) 0.544 (binomial cdf) (d) 0.968

7. (a) Mean 2, variance 1.8 (b) $E(X) = 9$, variance 3.6

8. (a) 0.282 (b) 0.876 (c) 2 (d) 0.205 (e) The trials are not independent and p is not constant

9. (a) $\frac{1}{9}$ (b) $k = \frac{1}{4}$, $P(X < 1) = \frac{1}{4}$ (c) $\frac{3}{4}$ (d) 2.67 (e) 0.794

10. (a) $\frac{3}{32}$ (b) Mode 2: f is a downward parabola with roots 0 and 4, so the maximum is halfway between (c) 2, by symmetry about $x = 2$ (or by integrating $\int_0^4 x f(x)\,dx$) (d) Area $= \frac{1}{2} + \frac{1}{2} = 1$; $P(X < 1.5) = 0.75$ (e) 1

11. (a) 0.841 (b) 0.0912 (c) 0.683 (d) 0.933 (e) 0.0228

12. (a) 0.0548 (b) ≈ 0.683, agreeing with the 68% rule (c) 43.8, so about 44 packets (d) $\mu \pm 2\sigma$, that is from 450 g to 550 g

13. (a) 2 (b) 78.2 (c) 90.4

14. (a) 54.2 (b) 35.0 (c) 14.8

15. (a) $z_A = 1$, $z_B \approx 1.33$; B is the better performance (b) $\mu = 27.5$, $\sigma \approx 5.85$ (c) $Q_1 \approx 44.6$, $Q_3 \approx 55.4$, IQR ≈ 10.8

END-OF-CHAPTER PROBLEMS

1. (a) 0.4 (b) 2.6 (c) 0.84

2. (a) \$1.10 (b) Loss of \$0.90 per ticket; the raffle favours the charity

3. (a) $X \sim B(15, 0.08)$ (b) 0.286 (c) 0.340 (d) 1.2

4. (a) $X \sim B(30, \frac{1}{6})$, mean 5 (b) 0.192 (c) 0.223

5. (a) $\frac{2}{9}$ (b) $\frac{4}{9}$ (c) 2 (d) Median $\frac{3}{\sqrt{2}} \approx 2.12$; mode 3

6. (a) 0.106 (b) 0.864 (c) 0.106, so about 10.6%

7. (a) 603 (b) 632

8. (a) $\frac{40-\mu}{\sigma} = -1.282$ and $\frac{70-\mu}{\sigma} = 0.6745$ (b) $\sigma \approx 15.3$, $\mu \approx 59.7$

9. (a) 9 (b) Variance 16, standard deviation 4

10. (a) 0.106 (b) 3.17 (c) 0.965

11. (a) $f \geq 0$ on $[0, 4]$ and $\int_0^4 f(x)\,dx = 1$, so f is valid (b) $\frac{3}{4}$ (c) 0

12. (a) $a = 0.3$, $b = 0.4$ (b) 0.625

13. (a) 2.60 dollars (b) \$2.60 (c) $-\$0.40$, so over 100 plays the expected loss is \$40

14. (a) A fixed number of independent trials ($n = 12$), each with two outcomes; a constant germination probability $p = 0.85$ (b) 0.292 (c) 0.736 (d) Mean 10.2, standard deviation 1.24

15. (a) 0.254 (b) 11

16. (a) 0.3 (b) 0.267 (c) $E(X) = 3$, variance 2.1

17. (a) 0.0912 (b) 0.656 (c) 912

18. (a) 188 g (b) 127 g (c) 0.87

19. (a) 4.82 min (b) 0.15: 20 and 30 are symmetric about the mean 25, so $P(X < 20) = P(X > 30)$

20. (a) $300 = \mu - 2.054\sigma$ and $340 = \mu + 1.645\sigma$ (b) $\sigma \approx 10.8$ g, $\mu \approx 322$ g

21. (a) Proof (b) Constant at height $\frac{2}{3}$ on $[0, 1]$, then a straight line falling from $\frac{2}{3}$ at $x = 1$ to 0 at $x = 2$; zero elsewhere (c) $\frac{1}{12}$ (d) $\frac{3}{4}$ (e) $\frac{7}{9}$

22. (a) Proof (b) $\frac{1}{2}$, a maximum (c) $\frac{1}{2}$ (d) $\frac{1}{20}$

23. (a) 0.841 (b) 0.401 (c) 0.756

24. (a) Proof (b) 0.263 (c) 0.254, close to the modelled 0.263; the normal model is a little high here (d) 0.668, or 66.8%, against the 68.3% the normal model predicts (e) A good fit, to within about one percentage point

CHALLENGE QUESTIONS

25. (a) W takes $20, 5, 0$ with probabilities $\frac{1}{36}, \frac{5}{36}, \frac{30}{36}$ (b) \$1.25 (c) \$1.25; fair means the expected gain is zero (d) \$0.75 per game, so \$375 over 500 games (e) Variance $\frac{625}{48} \approx 13.0$, standard deviation 3.61 (f) Chance dominates ten games; the stall's average approaches \$0.75

26. (a) $E(X) = 25$, variance 12.5; so $\mu = 25$, $\sigma \approx 3.54$ (b) 0.8811 (binomial cdf) (c) 0.8427, about 0.04 too small: it misses the outer halves of the bars at 20 and 30 (d) ≈ 0.8802, matching 0.8811 to two decimal places (e) Binomial 0.6563; unadjusted 0.4729; adjusted 0.6572 (f) Use the continuity correction; accuracy improves as n grows

27. (a) Proof (b) Proof (c) $(n+1)p = 6.3$, so the mode is 6 (d) Equality holds at $r = (n+1)p$, so there are two modes; for $B(9, 0.5)$, $P(X = 4) = P(X = 5) \approx 0.246$

Chapter 13 · Limits & Derivatives

YOUR TURN

1. (a) 5 (b) 5 (c) 0 (d) $\frac{1}{2}$

2. (a) $x + 4 \to 8$ (b) $x + 1 \to 2$ (c) $x + 1 \to 3$ (d) $x - 3 \to -6$

3. (a) 1 (b) Sketch: a central peak of height 1 at $x = 0$, decaying ripples crossing at $x = \pm\pi, \pm 2\pi, \pm 3\pi$ (c) Undefined at $x = 0$ (a hole), with the curve approaching 1; $y \to 0$ as $x \to \pm\infty$

4. The one-sided limits disagree, so there is no single limit

5. (a) Converges to 0 (b) Diverges: oscillates between -1 and 1 without approaching a value (c) Diverges: grows without bound (d) $3+\frac{1}{x}\to 3$: converges to 3

6. $x=0.1$: 0.49958; $x=0.01$: 0.50000 (5 d.p.). The limit appears to be $\frac{1}{2}$

7. (a) $2x+h\to 2x$ (b) $6x+3h\to 6x$ (c) $2x+h+1\to 2x+1$ (d) $2x+h-3\to 2x-3$

8. $h=0.1$: 0.99833; $h=0.01$: 0.99998; $h=0.001$: 1.00000. The quotient approaches 1, confirming $\frac{d}{dx}\sin x=\cos x$ at $x=0$, since $\cos 0=1$

9. At $x=0$: $1.0517,\ 1.0050,\ 1.0005$, approaching $1=e^0$. At $x=1$: $2.8588,\ 2.7319,\ 2.7196$, approaching $2.71828\ldots=e^1$. The gradient equals the value of the function at each point, suggesting $\frac{d}{dx}e^x=e^x$

10. $f'(x)$, $\frac{dy}{dx}$, $\frac{d}{dx}f(x)$

11. The difference quotient is $\frac{|h|}{h}$, which is 1 for $h>0$ and -1 for $h<0$; the one-sided limits disagree, so the limit defining $f'(0)$ does not exist. The graph has a corner: unbroken, but not smooth

12. (a) $7x^6$ (b) $15x^2-2$ (c) $4x^3-6x$ (d) $20x^4+1$ (e) $6x^2-14x+1$ (f) 0

13. (a) $2x^{-1}\Rightarrow -\frac{2}{x^2}$ (b) $x^{1/2}\Rightarrow \frac{1}{2\sqrt{x}}$ (c) $x^3+x^{-1}\Rightarrow 3x^2-\frac{1}{x^2}$ (d) $x^{-3}\Rightarrow -\frac{3}{x^4}$ (e) $x^{1/3}\Rightarrow \frac{1}{3\sqrt[3]{x^2}}$ (f) $x+x^{-1}\Rightarrow 1-\frac{1}{x^2}$

14. (a) $2x-5$, so 1 (b) $3x^2-4$, so -1 (c) $\frac{1}{2\sqrt{x}}$, so $\frac{1}{4}$ (d) $6x^2+1$, so 7

15. (a) 3 (b) ± 2

16. The power rule applies to the form x^n, and a root is not written that way; rewritten, $y=x^{1/4}$, giving $\frac{1}{4}x^{-3/4}$

17. (a) $4x-4$ (b) $-2x+2$ (c) $-x+2$ (d) $x+1$

18. (a) $y-1=-\frac{1}{3}(x-1)$ (b) $y-2=-4(x-4)$

19. They are perpendicular: $m_{\text{norm}}=-\frac{1}{m_{\text{tan}}}$. If the tangent is horizontal the normal is vertical, with equation $x=a$; its gradient is undefined

20. ± 2

21. -9, so $(3,-9)$

22. (a) $\cos x-\sin x$ (b) $e^x-\frac{3}{x}$ (c) $-4\sin x$ (d) $\sec^2 x$ (e) $\frac{1}{1+x^2}$ (f) $3^x\ln 3$

23. (a) $2\cos x-5e^x$ (b) $\frac{1}{x}+\sec^2 x$ (c) $\sec x\tan x$ (d) $\frac{1}{\sqrt{1-x^2}}$

24. $\frac{1}{2}$

25. $e^x=1\Rightarrow x=0$. At $(0,1)$ the curve has gradient 1; this is the point where the tangent is $y=x+1$

26. The standard derivatives rest on $\lim_{x\to 0}\frac{\sin x}{x}=1$, which holds in radians. In degrees that limit is $\frac{\pi}{180}$, not 1, so an extra constant factor would appear in every derivative

27. (a) $8(2x+1)^3$ (b) $2x\cos(x^2)$ (c) $3e^{3x}$ (d) $\frac{3}{2\sqrt{3x+1}}$

28. (a) $\cos x-x\sin x$ (b) $xe^x(x+2)$ (c) $x^2(3\ln x+1)$ (d) $\sin x+x\cos x$

29. (a) $\frac{1}{(x+1)^2}$ (b) $\frac{-2}{(x-1)^2}$ (c) $\frac{x\cos x-\sin x}{x^2}$ (d) $\frac{e^x(x-1)}{x^2}$

30. $u=v=x$: $(uv)'=2x$ but $u'v'=1$

31. 108

32. The derivative of the inner function has been omitted; correct: $3\cos(3x)$

33. (a) $6x-12$ (b) $4x^3$, then $12x^2$ (c) $\cos x$, then $-\sin x$ (d) $2e^{2x}$, then $4e^{2x}$

34. (a) $2x-4>0\Rightarrow x>2$ (b) Testing the four intervals: $-1<x<0$ or $x>1$ (c) $0\le x<\frac{\pi}{2}$ or $\frac{3\pi}{2}<x\le 2\pi$ (d) $x<1$

35. (a) $6x>0\Rightarrow x>0$ (b) $6x-6>0\Rightarrow x>1$ (c) $\frac{\pi}{2}<x<\frac{3\pi}{2}$ (d) None: $\ln x$ is concave down for all $x>0$

36. $(1,-2)$; f'' changes sign at $x=1$

37. Stationary points at the zeros of f': $x=-2$, where f' changes $+$ to $-$, a maximum; and $x=3$, where f' changes $-$ to $+$, a minimum

38. $v=6t^2-6t$, $a=12t-6$; $a=0$ at $t=\frac{1}{2}$ second

39. (a) $\frac{0}{0}$; $\frac{e^x}{1}\to 1$ (b) $\frac{0}{0}$; $\frac{2\cos 2x}{1}\to 2$ (c) $\frac{0}{0}$; $\frac{\sec^2 x}{1}\to 1$ (d) $\frac{\infty}{\infty}$; $\frac{1/x}{2x}=\frac{1}{2x^2}\to 0$

40. (a) $\frac{0}{0}\to\frac{\sin x}{2x}$, still $\frac{0}{0}\to\frac{\cos x}{2}\to\frac{1}{2}$ (b) $\frac{0}{0}\to\frac{e^x-1}{2x}$, still $\frac{0}{0}\to\frac{e^x}{2}\to\frac{1}{2}$

41. L'Hôpital: $\frac{\infty}{\infty}\to\frac{2x}{4x}=\frac{1}{2}$. Dividing by x^2: $\frac{1+1/x^2}{2-3/x^2}\to\frac{1}{2}$. Both give $\frac{1}{2}$

42. The limit is not indeterminate: direct substitution gives $\frac{2}{1}$, and L'Hôpital's rule applies only to $\frac{0}{0}$ and $\frac{\infty}{\infty}$. The correct value is 2

43. $\frac{0}{0}$ and $\frac{\infty}{\infty}$. A form such as $\frac{2}{0}$ is not indeterminate: it already tells you the quotient grows without bound, so differentiating would give a wrong answer

END-OF-CHAPTER PROBLEMS

1. $e^x(\sin x + \cos x)$; at $x = 0$ the gradient is 1

2. $6x - 9$

3. 2

4. (a) $3x^2$ (b) $\frac{1}{2\sqrt{x}}$ (c) e^x (d) $\cos x$

5. Proof

6. (a) $3x^2 - 3$ (b) ± 1. Points: $(1, -2)$ and $(-1, 2)$

7. (a) $f(-0.1) = 0.95163$, $f(-0.01) = 0.99502$, $f(0.01) = 1.0050$, $f(0.1) = 1.0517$ (b) 1 (c) 1

8. $6x - 2$

9. (a) V-shaped graph with vertex at $(2, 0)$ (b) As $x \to 2$ from either side, $|x - 2| \to 0 = f(2)$: no hole or jump (c) $\frac{|h|}{h}$ is 1 for $h > 0$ and -1 for $h < 0$; the one-sided limits disagree, so no derivative exists (a corner)

10. $\frac{9}{2}$

11. Points: $(2, 2)$ and $(-2, -2)$

12. (a) $\frac{2e^{2x}(x^2-x+1)}{(1+x^2)^2}$ (b) 2

13. $\frac{1}{6}$

14. (a) $2x - 2$ (b) 4

15. (a) 3, so the point is $(3, 3)$ (b) $y = 4x - 9$ (c) $y - 3 = -\frac{1}{4}(x - 3)$

16. (a) $\frac{1}{2}$ (b) $\frac{1}{2}$

17. (a) $\frac{1-\ln x}{x^2}$ (b) 0, so the tangent is horizontal

18. (a) $24x^2(2x^3 - 1)^3$ (b) $x(2\ln x + 1)$ (c) $\frac{x\cos x - \sin x}{x^2}$ (d) $e^{3x}(3\cos x - \sin x)$ (e) $\frac{4x}{\sqrt{4x^2+1}}$ (f) $\frac{e^x(x-1)^2}{(x^2+1)^2}$

19. (a) Proof (b) $y - e = -\frac{1}{e}(x - 1)$

20. (a) $\frac{3}{1+9x^2}$ (b) $2\sec(2x)\tan(2x)$ (c) $\frac{1}{x\ln 2}$ (d) $\frac{2x}{\sqrt{1-x^4}}$ (e) $-\operatorname{cosec} x \cot x$ (f) $-3\operatorname{cosec}^2(3x)$

21. (a) $3(x-1)(x-3) > 0 \Rightarrow x < 1$ or $x > 3$ (b) $6x - 12 > 0 \Rightarrow x > 2$ (c) 2

22. $a = \pm 1$, giving $y = 2x - 1$ and $y = -2x - 1$

23. (a) $30x(3x^2 + 1)^4$ (b) $e^x(x^2 + 2x)$ (c) $\frac{1}{(x+1)^2}$ (d) $\frac{6x}{3x^2+2}$ (e) $-\frac{x\sin x + 2\cos x}{x^3}$ (f) $7(x+1)^2(2x-5)^3(2x-1)$

24. (a) $\sec x \tan x$ (b) $-\tan x$ (c) $\arctan x + \frac{x}{1+x^2}$ (d) $\frac{-2}{\sqrt{1-4x^2}}$ (e) $\cos x\, e^{\sin x}$ (f) $\frac{x-(1+x^2)\arctan x}{x^2(1+x^2)}$

25. (a) $3t^2 - 12t + 9$ (b) $t = 1$ and $t = 3$ seconds (c) $6\ \mathrm{m\,s^{-2}}$

26. (a) $f'(x) > 0 \Rightarrow x < -2$ or $x > 4$ (b) $x = -2$ (maximum) and $x = 4$ (minimum) (c) 1

27. $\frac{1}{3}$

28. (a) $f'(x) = 3x^2 - 6x - 9 = 3(x-3)(x+1)$, zero at $x = -1$ and $x = 3$ (b) $f'(x) > 0$ for $x < -1$ or $x > 3$: increasing on these intervals (c) $f''(x) = 6x - 6$, so $x = 1$ (point of inflexion)

29. (a) $3(x-3)(x+1)$ (b) $(-1, 10)$ and $(3, -22)$ (c) $f''(x) = 6(x-1)$; $f''(-1) = -12 < 0$, a maximum, and $f''(3) = 12 > 0$, a minimum (d) $(1, -6)$ (e) Concave up for $x > 1$

30. (a) Vertical $x = -2$; horizontal $y = 2$ (ratio of leading coefficients) (b) Proof (c) $(x+2)^2 > 0$ for all $x \neq -2$, so $g'(x) > 0$ throughout the domain (d) $y = \frac{5}{4}x - \frac{1}{2}$ (e) $y = -\frac{4}{5}x - \frac{1}{2}$

31. (a) Proof (b) $(0, 0)$ and $(2, 4e^{-2}) \approx (2, 0.541)$ (c) Proof (d) At $x = 0$: $f'' = 2 > 0$, minimum. At $x = 2$: $f'' = -2e^{-2} < 0$, maximum (e) $y \to 0^+$ as $x \to \infty$; horizontal asymptote $y = 0$

32. (a) Proof (b) Proof (c) $4\pi r - \frac{1000}{r^2}$ (d) 4.30 cm (e) $4\pi + \frac{2000}{r^3} > 0$ for $r > 0$, so the stationary value is a minimum (f) Proof

CHALLENGE QUESTIONS

33. (a) $x^n + nx^{n-1}h + {}^nC_2 x^{n-2}h^2 + \cdots$ (b) Proof (c) Proof (d) $4x^3h + 6x^2h^2 + 4xh^3 + h^4$; dividing by h gives $4x^3 + 6x^2h + 4xh^2 + h^3 \to 4x^3$ (e) The binomial theorem in this form needs n to be a positive integer, so the expansion in (a) fails; directly, $\frac{1}{h}\left(\frac{1}{x+h} - \frac{1}{x}\right) = \frac{-1}{x(x+h)} \to -\frac{1}{x^2} = -x^{-2}$, which agrees

34. (a) 0.99833, 0.99998, 1.00000, and the same for negative x (b) Proof (c) Proof (d) Proof (e) In degrees $\lim_{x\to 0}\frac{\sin x}{x} = \frac{\pi}{180}$, not 1, so (d) would give $\frac{\pi}{180}\cos x$; the clean formula holds only in radians

35. (a) $\frac{1-\cos x}{3x^2}$, then $\frac{\sin x}{6x}$, then $\frac{\cos x}{6} \to \frac{1}{6}$: three applications, each checked indeterminate first (b) Dividing by x^3 gives $\frac{1}{6} - \frac{x^2}{120} + \cdots \to \frac{1}{6}$ (c) $\frac{x^2}{2} + \frac{x^3}{6} + \cdots$, so the fraction is $\frac{1}{2} + \frac{x}{6} + \cdots \to \frac{1}{2}$ (d) L'Hôpital is general; the series method is quicker here. A harder example: $\lim_{x\to\infty}\frac{\ln x}{x}$

YOUR TURN

1. (a) -4, so $(3, -4)$ (b) ± 2, giving $(2, -16)$ and $(-2, 16)$ (c) Points $(2, -19)$ and $(-1, 8)$ (d) Points $(0, 0)$, $(1, -1)$, $(-1, -1)$

2. (a) $2 > 0$, so $(3, -4)$ is a local minimum (b) At $x = 2$ it is $12 > 0$, a minimum; at $x = -2$ it is $-12 < 0$, a maximum (c) $2 > 0$, a local minimum at $(1, 2)$ (d) 4: minimum

3. (a) 0 is stationary (b) The test gives no conclusion when the second derivative is zero (c) Stationary point of inflexion

4. (a) $x = \ln 3$: minimum at $(\ln 3, 3 - 3\ln 3) \approx (1.10, -0.296)$ (b) $x = e^{-1/2}$: minimum at $\left(e^{-1/2}, -\frac{1}{2e}\right) \approx (0.607, -0.184)$ (c) $x = \frac{\pi}{4}$: maximum, $y = \sqrt{2}$; $x = \frac{5\pi}{4}$: minimum, $y = -\sqrt{2}$ (d) $x = \frac{1}{2}$: maximum at $\left(\frac{1}{2}, \frac{1}{2e}\right) \approx (0.5, 0.184)$

5. (a) -11, so $(2, -11)$ (b) 2, so $(-1, 2)$ (c) $\frac{9}{2}$, so $\left(\frac{3}{2}, \frac{9}{2}\right)$

6. (a) 0 is a minimum (b) This is a genuine point of inflexion (c) $6x - 12 < 0$ when $x < 2$, so the curve is concave down on $x < 2$

7. (a) $(-2, -2e^{-2}) \approx (-2, -0.271)$ (b) $x = \frac{\pi}{2}$ and $x = \frac{3\pi}{2}$ (c) $\frac{d^2y}{dx^2} = -\frac{1}{x^2} < 0$ for every $x > 0$, so the concavity never changes (d) $\frac{d^2y}{dx^2} = (x^2 - 1)e^{-x^2/2}$, zero and sign-changing at $x = \pm 1$

8. (a) 100 (b) The rectangle is a 10×10 square (c) 12

9. (a) Need $x > 0$ and $12 - 2x > 0$, so $0 < x < 6$ (b) 2 gives the maximum (c) 128 cm^3

10. (a) 1250 m^2 (b) $2\pi + \frac{4000}{r^3} > 0$, a minimum (c) 5

11. (a) $t = \frac{10}{3} \approx 3.33$ h; maximum $50e^{-1} \approx 18.4$ mg L^{-1} (b) $x = 1$, minimum value 1 (c) $x \tan x = 1$, so $x \approx 0.860$ and $A \approx 1.12$

12. (a) 2 m s^{-2} (b) 2 s (c) 6 m s^{-2} (d) 1 m s^{-2}

13. (a) 3 s (b) 6 m s^{-2} (c) The signs are the same, so the particle is speeding up (d) The signs differ, so the particle is slowing down

14. (a) $v = 6\cos 2t$, $a = -12\sin 2t = -4s$ (b) $t = \frac{\pi}{4} \approx 0.785$ s, $s = 3$ m (c) -3 m s^{-2}; $e^{-0.5t} > 0$ always, so v is never zero (d) $t = \frac{\pi}{6}$ and $t = \frac{5\pi}{6}$

15. (a) $-\frac{x}{y}$ (b) $-\frac{y}{x}$ (c) $\frac{x}{y}$ (d) $-\frac{x^2}{y^2}$

16. (a) $-\frac{2x+y}{x}$ (b) $-\frac{2xy}{x^2+2y}$

17. (a) $\frac{4}{3}$ (b) $4 - x$ (c) 25, giving $(0, 5)$ and $(0, -5)$ (d) Naming y as well selects which point on the curve is meant

18. (a) $\dfrac{2x}{e^y - 1}$ (b) $-\dfrac{2x}{\cos y}$ (c) $-\dfrac{y \ln y}{x}$ (d) -1

19. (a) 20 cm^{2}s^{-1} (b) 75.4 cm^{2}s^{-1} (c) 300 cm^{3}s^{-1} (d) 2.51 cm s^{-1}

20. (a) 0.0398 cm s^{-1} (b) $\frac{dy}{dt} = -1.5$ m s^{-1} (c) 0.127 cm s^{-1}

21. (a) $\frac{1}{30} \approx 0.0333$ rad s^{-1} (b) $\frac{40}{3} \approx 13.3$ m s^{-1} (c) $32e^{0.6} \approx 58.3$ per year

END-OF-CHAPTER PROBLEMS

1. Proof

2. 25.1 m^{2}s^{-1}

3. (a) $1, 3$ s (b) -3 m s^{-1} (minimum velocity)

4. 800 dollars

5. -2

6. (a) $-\frac{2x+y}{x+2y}$ (b) $-x + 2$

7. (a) $\frac{1}{3}\pi h^3$ (b) $\frac{dh}{dt} = -\frac{1}{8\pi} \approx -0.0398$ cm s^{-1}

8. (a) -2 (b) 1

9. (a) 4 (b) 1

10. (a) 15 (b) $\frac{2-x}{y+1}$ (c) $3 + \frac{1}{2}x$

11. 0.675 m s^{-1}

12. 0.212 m min^{-1}

13. 1.94

14. Proof

15. (a) $2\pi r^2 + \frac{1000}{r}$ (b) $4\pi + \frac{2000}{r^3} > 0$: minimum

16. (a) $2x + \frac{200}{x}$ (b) 20. ($L'' > 0$: minimum.) Dimensions 10×20 m

17. (a) $a = \frac{3}{2}$, $b = -6$ (b) $-9 < 0$ maximum

18. (a) $x^2 + \frac{128}{x}$ (b) 4 m. ($A'' > 0$: minimum.)

19. (a) Proof (b) 4000 cm^3

20. Proof

21. (a) -0.375 m s^{-1} (b) 0.4375 m^{2}s^{-1}

22. 4.24 units s^{-1}

23. (a) 3: min $(3, -22)$ (b) -6. Inflexion $(1, -6)$ (c) $f' < 0$ between the roots: decreasing on $-1 < x < 3$

24. (a) 3 s (b) -6 m s^{-2} (c) $-3 < 0$. Same sign, so the particle is speeding up

25. (a) $6 > 0$, min at $(3, 2)$ (b) $-3 \neq 0$: a non-stationary inflexion

26. (a) $2, 4$ s (b) 6 m s^{-2} (c) $2 < t < 3$ and $4 < t \le 6$

27. (a) 1: point $(1, \frac{1}{e})$ (b) $-e^{-1} < 0$: a local maximum (c) 2

28. (a) $4x^2(x-3)$ (b) -27, so $(0,0)$ and $(3,-27)$ (c) $36 > 0$: minimum (d) It is neither a maximum nor a minimum: a stationary inflexion (e) -16, so $(0,0)$ and $(2,-16)$

29. (a) $0 < x < 6$ (b) 2 (c) 6 is excluded by the domain (d) 128 cm^3

30. (a) $6t-12$ (b) 3 seconds (c) 0 m (d) -3 m s^{-1} (e) Same signs, so the particle is speeding up

31. (a) 72: the point lies on C (b) Proof (c) $\frac{4}{5}$ (d) $4x - 5y + 12 = 0$

32. (a) $(\ln 2,\ 2 - 2\ln 2) \approx (0.693, 0.614)$ (b) $f'' = 2 > 0$: a local minimum (c) $f''(x) = e^x > 0$ for every x, so the concavity never changes

33. (a) $v = 6\cos 3t$, $a = -18\sin 3t$ (b) Proof (c) $t = \frac{\pi}{6}$ and $t = \frac{\pi}{2}$ (d) 6 m s^{-1}, occurring at $s = 0$

34. (a) Proof (b) $(e, \frac{1}{e}) \approx (2.72, 0.368)$, a local maximum (c) $x = e^{3/2} \approx 4.48$

35. (a) $100\left[\cos\theta + \cos^2\theta - \sin^2\theta\right]$ (b) $\theta = \frac{\pi}{3}$ (c) $75\sqrt{3} \approx 130$ cm^2

36. (a) $1 \cdot e^0 + 0 = 1$: the point lies on C (b) Proof (c) $-\frac{1}{2}$ (d) $x + 2y = 1$

37. (a) $x = 2\tan\theta$; $\frac{d\theta}{dt} = 2\pi$ rad min^{-1} (b) Proof (c) $\frac{16\pi}{3} \approx 16.8$ km min^{-1} (d) $\sec^2\theta \to \infty$, so the speed grows without bound as the beam turns parallel to the shore

CHALLENGE QUESTIONS

38. (a) Proof (b) Proof (c) $4\pi + \frac{4V}{r^3} > 0$ (d) Proof (e) Waste in cutting the ends; the seam adds material

39. (a) Proof (b) Using D^2 avoids differentiating a square root (c) 1.94 (d) Proof (e) The shortest segment meets the curve along the normal

40. (a) A line has constant gradient, so it does not bend and $\kappa = 0$ (b) $\kappa(0) = 2$ (c) Far out the arms are steep but their *direction* changes very slowly (d) A circle looks the same at every point, so κ is constant (e) The circle of radius $\frac{1}{2}$ centred at $(0, \frac{1}{2})$ (f) $y'' = 2$ everywhere for $y = x^2$, yet $\kappa = 2/(1+4x^2)^{3/2}$ falls away (g) $y = x^3$ has $\kappa = 0$ at $x = 0$ but is bent elsewhere (h) Unequal axis scales are a vertical stretch, so apparent bending depends on the scales chosen

Chapter 15 · Integral Calculus

YOUR TURN

1. (a) $x^4 - 3x^2 + C$ (b) $\frac{x^3}{3} + \frac{3x^2}{2} - 5x + C$ (c) $-\frac{3}{x^2} + C$ (d) $\frac{2}{3}x^{3/2} + C$ (e) $\frac{x^3}{3} + \frac{1}{x} + C$ (f) $\frac{x^3}{3} - 2x + C$ (g) $2\sqrt{x} + C$ (h) $2x^4 + C$

2. (a) $2x^2 - 3x + 3$ (b) $2x^{3/2} + 4$ (c) $x^3 + x + 2$ (d) A point on the curve is needed to fix C (e) Two integrations introduce two arbitrary constants

3. (a) 6 (b) 12 (c) 1 (d) 2 (e) 1 (f) 6

4. (a) $\frac{26}{3}$ (b) $\frac{4}{3}$ (c) 2 (d) 0; the two halves cancel, while the area is 4 (e) 8

5. (a) 1.75 (b) 3.75 (c) $\frac{8}{3} \approx 2.67$ (d) 2.1875, closer to $\frac{8}{3}$ than the four-rectangle value 1.75 (e) Both converge to $\frac{8}{3}$

6. (a) $2\sin x + \cos x + C$ (b) $3e^x + \ln|x| + C$ (c) $\frac{1}{3}e^{3x} + C$ (d) $\frac{(3x-2)^5}{15} + C$ (e) $-\frac{1}{2}\cos 2x + C$ (f) $\frac{1}{4}\tan 4x + C$ (g) $\frac{1}{3}\sin(3x+1) + C$ (h) $\frac{1}{2}\ln|2x+5| + C$

7. (a) 3, so $\frac{1}{3}\arctan\frac{x}{3} + C$ (b) 4, so $\arcsin\frac{x}{4} + C$ (c) $\frac{5^x}{\ln 5} + C$ (d) $\frac{1}{3}\arctan\frac{x+2}{3} + C$ (e) $\frac{1}{2}\arcsin\frac{2x}{3} + C$ (f) $\arctan\frac{x}{2} + C$

8. (a) $\frac{1}{6}\ln\left|\frac{x-3}{x+3}\right| + C$ (b) $\ln|x+1| + 3\ln|x+2| + C$ (c) Factorises: partial fractions. No real roots: complete the square

9. (a) $x+3$, so the integral is $\frac{x^2}{2} + 3x + C$ (b) $2 + \frac{1}{x}$, so the integral is $2x + \ln|x| + C$ (c) $\frac{x^2}{2} + \frac{1}{x} + C$ (d) $\frac{1}{2}\arctan\frac{x+3}{2} + C$

10. (a) $\frac{(x^3+1)^5}{5} + C$ (b) $\frac{1}{2}e^{x^2} + C$ (c) $\frac{2}{3}(x^2+1)^{3/2} + C$ (d) $-\frac{1}{2(x^2+4)} + C$ (e) $\frac{1}{2}(\ln x)^2 + C$ (f) $\frac{1}{4}\sin^4 x + C$

11. (a) $\frac{31}{5}$ (b) $\frac{31}{5}$, the same (c) $\frac{1}{3}$

12. (a) $\cos x$: $x\sin x + \cos x + C$ (b) e^{2x}: $\frac{1}{2}xe^{2x} - \frac{1}{4}e^{2x} + C$ (c) x: $\frac{x^2}{2}\ln x - \frac{x^2}{4} + C$ (d) $x\arctan x - \frac{1}{2}\ln(1+x^2) + C$ (e) $\frac{x^3}{3}\ln x - \frac{x^3}{9} + C$ (f) $x\tan x + \ln|\cos x| + C$

13. (a) π (b) 1 (c) e^x would leave $\int \frac{x^2}{2}e^x\,dx$, with a higher power than before (d) $u = x^2$; $u = \arctan x$; $u = \sin x$ (e) Neither factor simplifies; the integral returns to itself

14. (a) $\frac{1}{6}$ (b) $\frac{32}{3}$ (c) $\frac{32}{3}$ (d) 9 (e) $\frac{8}{3}$

15. (a) 2 (b) $x+2$ is above the parabola between those values (c) $\frac{9}{2}$ (d) $-\frac{9}{2}$; the curves were subtracted the wrong way round

16. (a) 1 (b) $e^2 - 1 \approx 6.39$ (c) 2 (d) $e + \frac{1}{e} - 2 \approx 1.09$ (e) $\ln 4 \approx 1.39$

17. (a) $\frac{32\pi}{5}$ (b) 8π (c) 36π (d) $\frac{2\pi}{3}$ (e) $\frac{\pi^2}{2}$

18. (a) 8π (b) The two rotations sweep out different solids (c) $\frac{32\pi}{5}$

19. (a) $\frac{\pi^2}{4} \approx 2.47$ (b) $\frac{\pi}{2}(e^2-1) \approx 10.0$ (c) $\frac{3\pi}{8} \approx 1.18$ (d) $\frac{\pi^2}{4} \approx 2.47$ (e) Squaring a sine or cosine gives $\sin^2$ or $\cos^2$, which has no standard integral; squaring an exponential gives e^{2x}, which does

20. (a) 12 m (b) $3t^2+1$ (c) -3 m (d) 8 m (e) 0 m; the particle returns to its starting point (f) 23 m

21. (a) 3 m (b) $3-3e^{-2} \approx 2.59$ m (c) $v = 1-\cos t$ (d) Turns at $t=\frac{\pi}{2}$; distance 4 m

END-OF-CHAPTER PROBLEMS

1. (a) x^3-2x^2+x+2 (b) 4

2. (a) ± 2 (b) $\frac{32}{3}$

3. (a) 4. Points $(\pm 2, 4)$ (b) $\frac{64}{3}$

4. (a) $\frac{1}{2}\ln(x^2+4)+C$ (b) $\frac{1}{2}\ln 2$

5. (a) $\frac{1}{2}xe^{2x} - \frac{1}{4}e^{2x} + C$ (as in Your Turn Q12) (b) $\frac{1}{4}e^2 + \frac{1}{4}$

6. (a) $\displaystyle\int_0^4 \pi x\, dx$ (b) 8π

7. (a) $3t^2-4t+5$ (b) t^3-2t^2+5t (c) 10 m

8. (a) $\frac{16}{3}$ (b) 8π

9. (a) $\frac{1}{2}\arctan\frac{x}{2}+C$ (b) $\frac{\pi}{8}$

10. (a) $1, 3$ s (b) 8 m

11. (a) $2x^2-3x+3$ (b) y-intercept 3

12. (a) $\frac{1}{3}e^{3x}+C$ (b) $\frac{(4x-1)^7}{28}+C$ (c) $-\frac{1}{2}\cos 2x+C$

13. 7

14. (a) 1 (b) 2

15. $\frac{9}{2}$

16. 78

17. $\frac{1}{2}\ln 2$

18. $2\ln|x-1| - \ln|x+3| + C$

19. $\frac{1}{3}\arctan(\frac{x-2}{3})+C$

20. (a) $x\sin x + \cos x + C$ (b) $\frac{x^3}{3}\ln x - \frac{x^3}{9} + C$

21. (a) 8π (b) $\frac{32\pi}{5}$

22. (a) $\frac{4}{3}$ m (b) 4 m

23. (a) 8 (b) $1, 2, 3$ are $2, 5, 10$, so the estimate is 17 (c) $8 < 12 < 17$ (d) Increase, towards 12

24. (a) 1 (b) Region between them shaded (c) $\frac{9}{2}$ (d) The line is the upper boundary throughout

25. (a) $\frac{x^2}{2}\ln x - \frac{x^2}{4} + C$ (b) $\frac{e^2+1}{4}$ (c) $\frac{\pi}{2}-1$ (d) $\cos x$ would leave $\int \frac{x^2}{2}\sin x\, dx$, with a higher power of x than before

26. (a) $\frac{\pi}{7}$ (b) $\frac{3\pi}{5}$ (c) The second, $\frac{3\pi}{5}$; its region lies further from its axis

27. (a) $6t-t^2$ (b) 6 s (c) 9 m s^{-1} (d) 36 m (e) The velocity never changes sign on this interval

28. (a) $\frac{x^2}{2} - x + 2\ln|x+1| + C$ (b) $\ln 3$ (c) $\frac{1}{3}$ (d) $\frac{1}{3}\arctan\frac{x-2}{3}+C$

29. (a) $\tan x = 1$, so $x = \frac{\pi}{4}$ (b) $\sqrt{2}-1 \approx 0.414$ (c) Proof

30. (a) $\pi \displaystyle\int_0^{\pi/2} \cos^2 x\, dx$ (b) Proof (c) On this interval the sine graph is the cosine graph reflected in $x = \frac{\pi}{4}$, and reflecting does not change the volume

31. (a) $t = 0, \frac{\pi}{2}, \pi$ s (b) 0 m (c) 12 m (d) The particle reverses at $t=\frac{\pi}{2}$, so the return cancels in displacement but adds to distance

32. (a) Proof (b) $1+e^{-2} \approx 1.14$ m (c) $2\,\text{m}\,\text{s}^{-1}$; the acceleration dies away exponentially, so the velocity levels off

33. (a) $e^{2x}=1$, so $x=0$ (b) $e+\frac{1}{e}-2 \approx 1.09$ (c) $\pi\left(\frac{e^2+e^{-2}}{2}-1\right) \approx 8.68$

34. (a) Parts: $\frac{x}{3}e^{3x} - \frac{1}{9}e^{3x} + C$ (b) Double angle: $\frac{x}{2} + \frac{\sin 6x}{12} + C$ (c) Substitution: $-\ln|\cos x| + C$ (d) Substitution: $\ln\frac{3}{2} \approx 0.405$

CHALLENGE QUESTIONS

35. (a) A cylinder of radius r and height h (b) $\frac{1}{3}\pi r^2 h$ (c) $\frac{4}{3}\pi r^3$ (d) From integrating x^2 to give $\frac{x^3}{3}$

36. (a) $\frac{1}{2}e^x(\sin x + \cos x) + C$ (b) I, which is true but gives no information (c) $e^x(x^2-2x+2)+C$; the power drops each round (d) Four rounds

37. (a) 1 (b) $\frac{n-1}{n}I_{n-2}$ (c) $\frac{8}{15}$ (d) Even n traces back to $I_0 = \frac{\pi}{2}$, odd to $I_1 = 1$

Chapter 16 · Differential Equations

YOUR TURN

1. (a) $x^2 + C$ (b) Ae^x (c) $A(1+x) - 1$ (d) $\ln(e^x + C)$

2. (a) $2e^{x^2/2}$ (b) e^{x^3} (c) $4\sec x$ (d) One integration gives one constant; the initial condition fixes it

3. (a) $x(2\ln|x| + C)$ (b) $x(C - \ln|x|)$ (c) $x(Ax - 1)$ (d) Cx

4. (a) Proof (b) $x^2(2\ln|x| + C)$

5. (a) $4 + Ce^{-x}$ (b) $3 + Ce^{-2x}$ (c) $\frac{1}{3}e^x + Ce^{-2x}$ (d) $e^{2x} + Ce^x$

6. (a) x^3 (b) $\frac{x^2}{4} + \frac{C}{x^2}$ (c) $\frac{x^2}{3} + \frac{2}{3x}$ (d) It makes the left side $\frac{d}{dx}(Iy)$

7. (a) $1 + 2x + 2x^2$ (b) $1 - \frac{x^2}{2} + \frac{x^4}{24}$ (c) $2x - 2x^2 + \frac{8x^3}{3}$ (d) $1 - x^2 + \frac{x^4}{2}$

8. (a) $2x - \frac{4x^3}{3}$ (b) $x^2 - \frac{x^6}{6}$ (c) $2x - \frac{8x^3}{3}$ (d) $1 + \frac{x}{2}$

9. (a) 1.105 (b) 0.9801 (4 d.p.) (c) The cosine series omits nothing until degree 6; for e^x the first omitted term is degree 3 (d) $1 + x - \frac{x^3}{3}$ (e) $1 - \frac{x^2}{2} + \frac{x^4}{24}$, the series for $\cos x$ (f) $1 - x + x^2 - x^3 + \cdots$, the expansion of $\frac{1}{1+x}$

10. (a) $y'(0) = 1$ (b) $y''(0) = 2$, $y'''(0) = 2$ (c) $1 + x + x^2 + \frac{x^3}{3} + \cdots$ (d) Exact 1.3997; the series beats Euler's 1.362 on the same information

11. (a) $y(1) \approx 1.5$ (b) 1.21 (c) 1.48 (d) 1.221

12. (a) 2.4 (b) 0.25 (c) Euler gives 1.21, about 0.01 too small (d) $y = e^x$ is concave up, so every straight step lies below the curve

END-OF-CHAPTER PROBLEMS

1. (a) $500e^{0.2t}$ (b) 3695

2. (a) $25 + 65e^{-0.1t}$ (b) 48.9°C (c) As $t \to \infty$, $\theta \to 25$°C

3. (a) $\frac{1}{v} + v$ (b) $x^2(2\ln|x| + A)$

4. (a) x (b) $x^2 + \frac{1}{x}$

5. (a) 1.625 (b) 1.797 (c) Euler underestimates by about 0.17

6. (a) 1 (b) $1 - \frac{x^2}{2} + \frac{x^4}{24} - \cdots$ (c) 0.9801

7. (a) $x - \frac{x^3}{6} + \frac{x^5}{120} - \cdots$ (b) $\frac{1}{6} - \frac{x^2}{120} + \cdots \to \frac{1}{6}$

8. (a) $1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \cdots$ (b) $1 - x^2 + \frac{x^4}{2} - \cdots$ (c) 0.767

9. (a) $25 + 65e^{-kt}$ (b) 0.0486 (c) 30.2 minutes

10. (a) Proof (b) $\frac{x}{1-\ln x}$

11. (a) $\frac{x^3}{2} + \frac{x}{2}$ (b) $e^{-x} - e^{-2x}$

12. $\sin x + 2\cos x$

13. (a) Proof (b) 2.75 (time units)

14. (a) 1.55 (3 sf) (b) Below the true value

15. (a) 6.82 m s^{-1} (b) 7 m s^{-1}. Euler overshoots 7 then corrects back towards it

16. (a) $2x - 2x^2 + \frac{8x^3}{3}$ (b) $2x - 2x^2 + \frac{8x^3}{3}$

17. (a) $x - x^2 + \frac{x^3}{3} + \cdots$ (b) 0.0903330; the two agree to six decimal places

18. (a) $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots$ (b) 2.90 (c) Very slow: four terms give barely one correct digit

19. Divide by x^3: $\frac{1}{6} - \frac{x^2}{120} + \cdots \to \frac{1}{6}$

20. $1 + x + \frac{3x^2}{2} + \frac{4x^3}{3} + \cdots$

21. (a) e^{kt} (b) Proof (c) 14.0 mg L^{-1} (d) 20 mg L^{-1}; the concentration plateaus rather than accumulating (e) 7.68 hours (f) 2.31 hours

22. (a) $x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$ (b) $-x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \cdots$ (c) Proof; $2x + \frac{2x^3}{3} + \frac{2x^5}{5} + \cdots$ (d) $x = \frac{1}{2}$, giving ≈ 1.0958 against the true $\ln 3 \approx 1.0986$ (e) $\frac{1}{2}$, comfortably inside the valid range, and its terms shrink quickly

23. (a) Proof (b) Proof (c) 12 minutes (d) 0.36 m (e) 0.149 m, so the estimate is 41% too low (f) A decreasing, concave-up parabola; tangent steps fall below it

CHALLENGE QUESTIONS

24. (a) $20 + 60e^{-kt}$ (b) 0.0405 (c) $46.7°$ (d) Equivalently $60e^{-kt} > 0$ (e) 10 minutes, independent of the starting temperature

25. (a) $(1+h)^n$ (b) $(1 + \frac{1}{n})^n$ (c) $n = 2$: 2.25; $n = 4$: 2.4414; $n = 8$: 2.5658; $n = 16$: 2.6379 (exact $e = 2.71828$) (d) The error halves; Euler is a first-order method (e) e; Euler's estimate converges to the exact value

26. (a) $-k\sqrt{1+v^2}$ (b) x^{-k} (c) $\frac{1}{2}\left(x^{1-k} - x^{1+k}\right)$ (d) $(0,0)$, the dock itself (e) $k = 1$: lands $\frac{1}{2}$ upstream; $k > 1$: swept away

Chapter 17 · Complex Numbers

YOUR TURN

1. (a) $4-2i$ (b) $3-4i$ (c) $5+i$ (d) 5

2. (a) $-i$ (b) 1 (c) $-i$ (d) 1

3. (a) $\mathrm{Re}(z)=7$, $\mathrm{Im}(z)=-4$ (b) $7+4i$ (c) 65 (d) 14

4. (a) $1+2i$ (b) $\frac{1}{2}-\frac{1}{2}i$ (c) $-3-2i$

5. (a) $2+2i$ (b) $3+2i$

6. (a) 5 (b) 13 (c) $\sqrt{2}$ (d) 2

7. (a) $\frac{\pi}{4}$ (b) $\frac{\pi}{2}$ (c) π (d) $-\frac{3\pi}{4}$

8. (a) $2\operatorname{cis}\frac{\pi}{2}=2e^{i\pi/2}$ (b) $3\operatorname{cis}\pi=3e^{i\pi}$ (c) $\sqrt{2}\operatorname{cis}(-\frac{\pi}{4})=\sqrt{2}\,e^{-i\pi/4}$ (d) $2\operatorname{cis}\frac{2\pi}{3}=2e^{2i\pi/3}$

9. (a) $2+2\sqrt{3}\,i$ (b) $\sqrt{2}-\sqrt{2}\,i$ (c) -3 (d) i

10. (a) $2\sqrt{5}$ (b) The distance between the points representing z and w (c) Both equal 25

11. (a) $6\operatorname{cis}\frac{\pi}{2}$ (b) $3\operatorname{cis}\frac{\pi}{3}$ (c) $\operatorname{cis}\pi=-1$ (d) $8\operatorname{cis}\frac{\pi}{2}$

12. (a) 12 (b) $\frac{5\pi}{12}$ (c) $\frac{3}{4}$ (d) $\frac{\pi}{12}$

13. (a) $\cos 4\theta+i\sin 4\theta$ (b) $-8i$ (c) $32\operatorname{cis}\frac{\pi}{3}=16+16\sqrt{3}\,i$

14. (a) $2\cos\theta$ (b) $2\cos 3\theta$ (c) $2i\sin 2\theta$

15. (a) $3i$ (or $-3i$) (b) 3 (c) Five (d) A regular n-gon centred at the origin

16. (a) $1,\ -\frac{1}{2}+\frac{\sqrt{3}}{2}i,\ -\frac{1}{2}-\frac{\sqrt{3}}{2}i$ (b) $1,\ i,\ -1,\ -i$ (c) $\operatorname{cis}\frac{k\pi}{3}$, $k=0,\dots,5$ (d) Proof

17. (a) Modulus 2; arguments $0,\ \frac{2\pi}{3},\ \frac{4\pi}{3}$ (b) Modulus 2; arguments $0,\ \frac{\pi}{2},\ \pi,\ \frac{3\pi}{2}$ (c) Modulus 1; arguments $\frac{\pi}{6},\ \frac{5\pi}{6},\ \frac{3\pi}{2}$

18. (a) $\pm(3+2i)$ (b) $\pm(1+3i)$ (c) $\pm(2+3i)$

19. (a) $\pm 3i$ (b) $\pm 2i$ (c) $1\pm 2i$ (d) $-1\pm i$

20. (a) $3-i$ (b) z^2+4 (c) z^2-2z+2

21. (a) $b=-2$, $c=5$ (b) Non-real roots come in conjugate pairs, so three roots cannot all be non-real

22. (a) Proof (b) $\sin 2\theta=2\sin\theta\cos\theta$ (c) Proof (d) $\tan 2\theta=\dfrac{2\tan\theta}{1-\tan^2\theta}$

23. (a) $\cos^2\theta=\frac{1}{2}(1+\cos 2\theta)$ (b) $\sin^2\theta=\frac{1}{2}(1-\cos 2\theta)$ (c) $\cos^3\theta=\frac{1}{4}(\cos 3\theta+3\cos\theta)$

24. (a) Proof (b) $\cos^5\theta=\frac{1}{16}(\cos 5\theta+5\cos 3\theta+10\cos\theta)$

END-OF-CHAPTER PROBLEMS

1. (a) $4+2i$ (b) $11-2i$ (c) $-1+2i$ (d) 5

2. $\frac{1}{2}+\frac{1}{2}i$

3. (a) $|z|=\sqrt{2}$, $\arg z=\frac{3\pi}{4}$ (b) $|z|=2$, $\arg z=-\frac{\pi}{6}$

4. $2e^{i\pi/3}$

5. (a) 64 (b) $32i$ (c) -1

6. (a) $\pm(2+i)$ (b) $\pm(2+3i)$

7. (a) $4\operatorname{cis}\frac{2\pi}{3}$ (b) -8

8. $b=-2$, $c=5$

9. (a) $2\operatorname{cis}\frac{\pi}{6},\ 2\operatorname{cis}\frac{5\pi}{6},\ 2\operatorname{cis}(-\frac{\pi}{2})$ (b) $\sqrt{3}+i,\ \ -\sqrt{3}+i,\ \ -2i$

10. (a) -1 (b) i

11. $3\operatorname{cis}0,\ 3\operatorname{cis}\frac{2\pi}{3},\ 3\operatorname{cis}(-\frac{2\pi}{3})$

12. $x=1$, $y=1$

13. (a) $8e^{i\pi}$ (b) $\sqrt{3}+i$

14. (a) $\sqrt{2}+\sqrt{2}i,\ -\sqrt{2}+\sqrt{2}i,\ -\sqrt{2}-\sqrt{2}i,\ \sqrt{2}-\sqrt{2}i$ (b) Proof

15. $-\frac{1}{2}\pm\frac{3}{2}i$

16. Proof

17. A square; area 2

18. $z^2-4z+13$

19. $\pm(1+i)$

20. $|z|=2\sqrt{2}$, $\arg z=\frac{\pi}{4}$, so $z=2\sqrt{2}\,e^{i\pi/4}$

21. $3+4i$

22. Proof

23. Proof

24. Proof

25. $w=-8$; the other two cube roots are -2 and $1-\sqrt{3}\,i$

26. (a) $1,\ \operatorname{cis}\frac{2\pi}{5},\ \operatorname{cis}\frac{4\pi}{5},\ \operatorname{cis}\frac{6\pi}{5},\ \operatorname{cis}\frac{8\pi}{5}$ (b) Proof (c) Proof (d) Proof (e) Proof

27. (a) Proof (b) Proof (c) Proof (d) $\frac{3\pi}{16}$ (e) It turned a fourth power into easily integrated multiple angles

28. (a) The coefficients are real, so non-real roots occur in conjugate pairs (b) z^2-2z+5 (c) The other factor is z^2+9 (d) All four roots: $1+2i,\ 1-2i,\ 3i,\ -3i$ (e) Symmetric about the real axis; $P(z)$ has real coefficients

CHALLENGE QUESTIONS

29. (a) $|Z| = 50\,\Omega$, $\arg Z = 0.927$ rad (53.1°) (b) $200 \operatorname{cis} 0.927$ volts (c) $|Z|$ is the ratio of voltage to current amplitude; $\arg Z$ is how far the voltage peak leads the current peak (d) Proof (e) $\omega = 1000$ rad s^{-1}; both reactances equal $50\,\Omega$ (f) $|Z| = \sqrt{R^2 + X^2}$ is least when $X = 0$, giving the largest current; tuning moves resonance onto one station's frequency

30. (a) Proof (b) Proof (c) Amplitude $3\sqrt{3} \approx 5.20$, phase $\frac{\pi}{6}$ (d) Proof (e) $\phi = 0$: amplitude $2A$, the waves reinforce; $\phi = \pi$: amplitude 0, they cancel, which is how noise cancelling works

31. (a) Proof; the image is the segment $[-2, 2]$ (b) Proof (c) $\frac{5}{2}$ and $\frac{3}{2}$ (d) The height $r - \frac{1}{r} \to 0$ while the width $\to 2$, so the ellipse flattens onto the segment of part (a) (e) $\frac{dw}{dz} = 1 - \frac{1}{z^2}$ vanishes at $z = 1$, so angles are not preserved there and the curve comes to a point

32. (a) $1,\ i,\ -1,\ -i$; they sum to 0 (b) $X_0 = 4$, $X_1 = X_2 = X_3 = 0$ (c) For $k \neq 0$ each X_k sums all four roots of unity, which is 0; X_0 is the total, so it measures the constant level (d) $X_0 = 0$, $X_1 = 2$, $X_2 = 0$, $X_3 = 2$ (e) One cycle across the four samples, so frequency $k = 1$; X_3 is its mirror image, and $X_0 = 0$ says the signal averages zero

33. (a) $i,\ -1+i,\ -i,\ -1+i$; period 2 from z_2, so $c = i$ **is** in the set (b) $1,\ 2,\ 5,\ 26$; the terms escape (c) Proof (d) Each step increases the modulus, so it never returns; the pixel can be settled as soon as $|z_n| > 2$ (e) If $|c| > 2$ then $|z_1| = |c| > 2$, so it escapes; hence the set lies inside $|c| \leq 2$